INFORMATION CIRCULATION AND EXTERNAL-FLOW SUPPRESSION UNDER FINITE CAPACITY

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We study ensemble-averaged information measures in discrete-time stochastic systems whose components have finite state spaces. Their temporal evolution is induced by changes in the underlying joint probability distributions, as occur during relaxation or after an interaction is activated. We first derive the entropy and transfer-entropy balance identities needed for the analysis. Under deterministic reversible dynamics, specified restrictions on synergistic interactions, strong blocking flows, and fixed nonempty information-flow support, a directed cycle is necessary for the vector of component entropies to be constant or periodic. We then consider an external input merging with a pre-existing circulating input at a receiving node with q_P states and entropy capacity $C_P = \ln q_P$. Under explicit assumptions on the receiver, the external transfer entropy is bounded by the remaining conditional-entropy margin, $C_P - H(P'|P)$. The result constrains admissible transfer; a decrease in realized transfer is established separately for particular models. A two-bit network with a Fredkin-type gate attains the bound when the external interaction is activated and subsequently exhibits transient information dynamics. The analysis identifies how finite receiving capacity constrains additional information transfer in systems with pre-existing circulation.

Keywords: information circulation; information dynamics; transfer entropy; cross-talk.

1. Introduction

When several signals enter a component with finitely many possible states, the receiving component cannot retain arbitrarily much information about all of them. This raises a concrete question for a system with feedback: how much information can a new external input transfer to a component that already participates in an internal information circulation? We address this question through the evolution of Shannon entropy, mutual information, and transfer entropy at a specified temporal resolution.

Feedback loops occur in gene regulatory networks [1,2,12,17,42], thalamocortical circuits [45], Arctic climate feedbacks [25], and predator-prey systems [7]. Transfer entropy provides a common statistical language for describing directed predictive dependence in such systems [8, 44]. Here, *information circulation* means a directed cycle of nonzero transfer-entropy links between components.

Information dynamics is studied in information thermodynamics [19, 23, 24, 38, 49], quantum thermodynamics [13, 31, 41], neuroscience [47, 52], and bioinformatics [26]. Mutual information quantifies statistical dependence between components,

while transfer entropy quantifies the additional information that a source provides about a receiver's next state beyond knowledge of the receiver's present state. Their time dependence describes how these statistical and predictive relationships change as the underlying joint probability laws evolve, for example during relaxation or after an interaction is activated [3, 21, 40, 43].

The present work concerns information dynamics described by ensemble-averaged quantities for finite-state stochastic processes, without imposing strict stationarity. In practice, stationarity is often assumed when pooling observations over time, and stationary information measures have provided insights into ordering in kinetic Ising models [4]. Under strict stationarity, these ensemble quantities are constant, although transfer entropy can remain nonzero. Nonstationary behavior is common in complex systems and can be investigated through repeated-trial ensembles and suitably chosen analysis windows [16, 55, 56]. Here, activating an external input changes the joint distribution and initiates a transient response. This ensemble-averaged approach is distinct from the evolution of local information values along individual trajectories [32].

To describe this ensemble-level evolution, Ref. [30] developed an information dynamical framework that expresses finite-time changes in entropy, mutual information, transfer entropy, and reversed transfer entropy as exact balance identities. The additional terms, denoted by α, are defined combinations of conditional entropies and information measures. Their vanishing defines a restricted class of processes rather than an automatic property of an information measure. Section 2 introduces the process, defines the notation, and derives the pairwise balances used in this work.

We obtain two results with distinct assumptions. First, Sec. 3 shows that an active finite information-flow network must contain a directed cycle if all component entropies are constant or periodic, provided the deterministic reversible, blocking-flow, non-synergy, and fixed-support hypotheses hold. Section 4 then describes a single circulation and its response to an external input under specified zero-control conditions.

Second, Sec. 5 analyzes the receiving node directly. Its finite state space gives an entropy ceiling, and under the stated receiver assumptions the remaining conditional-entropy margin bounds the newly incoming transfer entropy. This is a bound on admissible transfer. Monotonic suppression of the realized transfer requires a specified family of probability laws and is examined separately in the examples. Section 6 gives a two-bit Fredkin-type construction and distinguishes the comparison across input ensembles from the subsequent evolution in time. The scope and assumptions are kept explicit throughout the calculations.

2. Stochastic Setting and Information Balances

2.1. *Finite-state nonstationary processes*

Consider an N-component stochastic process $\mathbf{\Gamma}(t) = (\Gamma_1(t), \ldots, \Gamma_N(t))$ at integer times t. Component i takes values in a fixed finite alphabet $\mathcal{M}_i$, with $q_i = |\mathcal{M}_i|$. The joint state space is $\mathcal{M} = \prod_i \mathcal{M}_i$. Thus the basic objects are the joint probability laws of a discrete-time, finite-state process. Deterministic dynamics are included by allowing the probability law to describe uncertainty over initial conditions in an ensemble.

Stationarity provides a useful statistical description when a system has reached an invariant regime. For example, stationary transfer-entropy analysis distinguishes the behavior of global information flow from that of mutual information near the ordering transition of a kinetic Ising model [4]. Such results motivate information-theoretic descriptions of complex systems, but they do not remove the need to describe transient or externally perturbed regimes.

For example, we can consider a Markov chain with a time-independent transition kernel and a non-invariant initial ensemble. Its probability law can evolve even though the transition kernel is fixed, changing the information shared between components. An external input can also change the transition kernel at a specified time and initiate a transient response. The interaction protocol in this paper starts from a stationary circulation, activates an external input, and examines the resulting transient. Allowing nonstationarity is therefore necessary to describe the response to that protocol. Nonstationary coupling has also been studied through time-resolved ensemble estimates [16], including applications to neural recordings [55].

Strict stationarity means that all finite-time joint laws are invariant under a common time shift. In that case, the ensemble entropies and transfer entropies used here are constant, although nonzero transfer can persist. We distinguish a stationary *information state*, meaning constancy of the specified information measures, from strict stationarity of the full process. Temporal fluctuations of local information values are a further, distinct description, introduced for comparison in Sec. 2.2.

The theory concerns distributions rather than a particular estimation procedure. When independent realizations are prepared under the same initial law and protocol, a two-time law can be estimated from observations at the same times $t, t+1$ across realizations. This does not require stationarity across t. Ensemble estimators exploit precisely this distinction between averaging over realizations and pooling over time [16]. The simulations in Sec. 6 use independent realizations and estimate distributions separately at each time.

For a single long trajectory, a common practical approach is to move an analysis window through time and reconstruct distributions within each window, assuming approximate stationarity over that interval. Sliding-window effective-transfer-entropy analysis provides an example [56]. The duration and position of the window are adjustable analysis choices: shorter windows improve temporal resolution but reduce the sample available for distribution estimation, whereas longer windows can

mix distinct regimes. Window size is also relevant when samples are pooled across repeated trials [55]. A window does not establish global stationarity or remove the need to justify local stationarity, adequate sampling, and estimator performance.

These estimation issues concern how the relevant laws are reconstructed from data. They do not preclude a theoretical description of how information functionals of time-dependent laws evolve. Conversely, the exact balance identities below do not by themselves provide an estimator for arbitrary nonstationary data or establish the accuracy of a windowed estimate. Their purpose is to analyze information dynamics once a consistent stochastic model or an adequately reconstructed family of distributions is given.

2.2. *One-step information measures and entropy capacity*

At a given base time, write $X = X(t)$, $X' = X(t+1)$, and $X'' = X(t+2)$.

Definition 1. For finite-state variables, Shannon entropy and conditional entropy are [9, 46]

$$\begin{aligned} H(X) &= -\sum_{x\in\mathcal{M}_X} p(x)\ln p(x), \\ H(Y|X) &= H(X,Y) - H(X). \end{aligned} \tag{1}$$

All logarithms are natural, so information is measured in nats, and $0\ln 0 = 0$. We also allow finite sets of variables as arguments. For $V = \{V_1, \ldots, V_m\}$, let p_V be the joint law on $\mathcal{M}_V := \prod_{j=1}^{m} \mathcal{M}_{V_j}$, with any fixed ordering of the variables. Then

$$\begin{aligned} H(V) := H(V_1, \ldots, V_m) &= -\sum_{\boldsymbol{v}\in\mathcal{M}_V} p_V(\boldsymbol{v})\ln p_V(\boldsymbol{v}), \\ H(V|W) &:= H(V\cup W) - H(W). \end{aligned} \tag{2}$$

Here W is a finite collection of conditioning variables, $H(\varnothing) = 0$, and a single variable is identified with its singleton collection. These conventions also apply to the set arguments in the co-information definition.

Definition 2. Mutual information and conditional mutual information are [9]

$$\begin{aligned} I(X,Y) &= H(X) + H(Y) - H(X,Y), \\ I(X,Y|Z) &= H(X|Z) + H(Y|Z) - H(X,Y|Z). \end{aligned} \tag{3}$$

Here the comma separates two arguments, each of which may be a random vector.

Definition 3. The one-step transfer entropy and reversed transfer entropy are

$$\begin{aligned} T_{Y\to X} &:= I(X', Y|X), \\ T^{\dagger}_{Y\to X} &:= I(X, Y'|X'). \end{aligned} \tag{4}$$

The first measures the additional predictive information provided by the present source Y about the next receiver state X', beyond the present receiver state X [8,44]. The second uses reversed time indexing [22, 48, 49].

Both are nonnegative conditional mutual informations. We consistently use one-step conditioning; for longer-memory processes this is a specified finite-history quantity, not a claim that conditioning on one receiver state captures its entire past. The reversed measure does not require a physically reversed experiment. Statistical transfer may also reflect common inputs or indirect paths, which is why structural update dependencies are specified separately below.

A prime on an information measure or coefficient shifts *every* time argument by one step. Thus

$$\begin{aligned} \Delta H(X) &= H(X') - H(X), \\ \Delta T_{Y\to X} &= T_{Y'\to X'} - T_{Y\to X}, \\ \Delta H(Y|X,X') &= H(Y'|X',X'') - H(Y|X,X'). \end{aligned} \tag{5}$$

The same rule defines $\Delta^2 F = F'' - 2F' + F$. When needed, $T_{Y\to X}(t)$ makes the base time explicit. Around activation, t_0, t_1, t_2 denote successive times, with $t_0 = t_1 - 1$ and $t_2 = t_1 + 1$.

For comparison with trajectory-level descriptions, the local one-step transfer entropy at a realized triple (x, x', y) is [32]

$$\tau_{Y\to X}(x, x', y; t) = \ln \frac{p_t(x'|x, y)}{p_t(x'|x)}, \qquad T_{Y\to X}(t) = \mathbb{E}_t[\tau_{Y\to X}]. \tag{6}$$

Here p_t denotes the relevant two-time law. A local value can fluctuate along a trajectory even when its expectation is stationary. All subsequent H, I, T, and $T^\dagger$ denote ensemble quantities, rather than these local values.

For a variable X with q_X states, we define its *alphabet entropy capacity* as

$$C_X := \max_{p \text{ on } \mathcal{M}_X} H_p(X) = \ln q_X. \tag{7}$$

In particular,

$$0 \le H(X'|X) \le H(X') \le C_X. \tag{8}$$

This capacity is the maximum over distributions on the chosen alphabet. A particular transition rule can impose a lower attainable maximum. The conditional entropy $H(X'|X)$ is the uncertainty in the next state after observing the present state; it is not itself a channel capacity optimized over inputs.

Finite capacity is an assumption on the modeled variables. It can arise from intrinsically discrete states, as in the logic-gate examples, or from a fixed finite observation partition of a continuous state space. In the latter case, the results concern the partition labels and depend on that partition. Coarse graining need not preserve Markovianity, determinism, or reversibility, so the structural hypotheses must be checked for the observed process. We do not apply the finite-alphabet bound to unpartitioned continuous variables or replace Shannon entropy by differential entropy. In particular, continuous state spaces do not by themselves imply infinite differential entropy; its behavior depends on the admissible distributions and constraints.

2.3. *Signed multiple-transfer terms and block transfer*

Ordinary transfer entropy already applies to groups of variables. For disjoint collections V and U, let $\mathbf{V}$ and $\mathbf{U}$ be fixed tuples containing their elements. Their ordinary block transfer entropy is

$$T_{\mathbf{V}\to\mathbf{U}} := I(\mathbf{U}', \mathbf{V}|\mathbf{U}) \geq 0. \tag{9}$$

This quantity measures total predictive information between two blocks. The separate multiple-transfer terms below are used to express exact corrections when a block contribution is decomposed into contributions from individual components.

Definition 4. For a nonempty finite collection V, co-information and conditional co-information are the alternating sums [5, 14, 18, 28, 35, 36, 50, 57]

$$\begin{aligned} I(V) &:= \sum_{\varnothing\neq S\subseteq V} (-1)^{|S|+1} H(S), \\ I(V|W) &:= \sum_{\varnothing\neq S\subseteq V} (-1)^{|S|+1} H(S|W). \end{aligned} \tag{10}$$

In particular, $I(\{X\}) = H(X)$ and $I(\{X, Y\}|W) = I(X, Y|W)$. Braces specify that the entries are separate co-information arguments; a bold tuple specifies one block argument.

Definition 5. For disjoint nonempty collections V, U, define the multiple transfer entropy and multiple reversed transfer entropy [30] by

$$\begin{aligned} T_{V\to U} &:= I(V \cup U'|U), \\ T^{\dagger}_{V\to U} &:= I(V' \cup U|U'), \qquad U' := \{X' : X \in U\}. \end{aligned} \tag{11}$$

The I on the right is conditional co-information, with one argument for each element of the union. Singleton braces may be omitted in subscripts.

For two sources and one receiver, this definition gives

$$\begin{aligned} T_{\{A,B\}\to P} &= I(A, P'|P) - I(A, P'|P, B), \\ T_{(A,B)\to P} &= T_{A\to P} + T_{B\to P} - T_{\{A,B\}\to P}. \end{aligned} \tag{12}$$

Thus the multiple-transfer term corrects the sum of pairwise transfers to recover the block transfer. It can be positive, zero, or negative. For independent fair-bit inputs to an XOR receiver, independent also of its present state, the two pairwise transfers vanish, the block transfer is $\ln 2$, and $T_{\{A,B\}\to P} = -\ln 2$. Negative multiple transfer therefore does not contradict nonnegativity of ordinary transfer entropy. Co-information combines contributions associated with redundancy and synergy; it is not used here as a nonnegative decomposition of either. Appendix A gives the dynamical XOR construction.

More generally, inclusion–exclusion yields

$$T_{\mathbf{K}\to X} = \sum_{\varnothing\neq S\subseteq K} (-1)^{|S|+1} T_{S\to X}, \tag{13}$$

and the same identity holds for reversed transfer. This identity, rather than a new definition of total block transfer, is the reason for introducing the signed terms. Define their net correction by

$$E_{K\to X} := \sum_{\substack{S\subseteq K\\ |S|\geq 2}} (-1)^{|S|}[T_{S\to X} - T^{\dagger}_{S\to X}]. \tag{14}$$

It follows that $T_{\mathbf{K}\to X} - T^{\dagger}_{\mathbf{K}\to X} = \sum_{k\in K}(T_{k\to X} - T^{\dagger}_{k\to X}) - E_{K\to X}$. An empty sum is zero; E need not have a fixed sign.

The utility of these signed quantities is also recursive. For a nonempty collection S, a further variable $Z \notin S$, and a conditioning collection W, the entropy definition gives

$$\begin{aligned} I(S\cup\{Z\}|W) &= I(S|W) - I(S|W,Z),\\ \Delta I(S\cup\{Z\}|W) &= \Delta I(S|W) - \Delta I(S|W,Z). \end{aligned} \tag{15}$$

Every argument, including the conditioning variables, is shifted in the second line. Thus a relation for one more co-information argument can be expressed through lower-cardinality co-informations with different conditioning. Together with the higher-order balances in Appendix B, this structure supports inductive arguments for controlling higher-order terms, in the sense of tracking their temporal changes or establishing cancellations under specified residual conditions. The cancellation of ΔE in Appendix C is one such use. Recursion alone neither closes the hierarchy nor forces higher-order correlations to vanish.

Partial information decomposition (PID) addresses the interpretation of information shared by multiple sources through redundant, unique, and synergistic contributions [53]. This distinction is valuable for interpreting and estimating multivariate information; PID has also been applied to information transfer [54]. The present analysis instead selects quantities for their exact entropy identities and recursive role in the time evolution of information measures. It does not require choosing a PID redundancy functional, and it does not identify a signed co-information with an individual PID contribution. The two approaches can therefore serve complementary questions about the same stochastic system.

2.4. *Derivation of the pairwise information balances*

Let K_i be a fixed collection of other component variables, with $K_i' = \{k' : k \in K_i\}$. The entropy identities below hold for any such collection. Later, the network analysis selects collections that contain the forward and inverse update dependencies. They are called sets of dynamical neighbors in that structural sense; this terminology does not infer causation from transfer entropy.

We first derive the entropy balance directly. Applying the entropy chain rule

twice gives

$$\begin{aligned}\Delta H(X) &= H(X'|X) - H(X|X') \\ &= T_{\mathbf{K}\to X} - T^{\dagger}_{\mathbf{K}\to X} + \alpha_1^{K\to X}, \\ \alpha_1^{K\to X} &:= H(X'|X,K) - H(X|X',K').\end{aligned} \tag{16}$$

Equation (13) then expresses the block transfers in terms of pairwise transfers and E. No stationarity or dynamical approximation has entered this step.

For a pair of components, expansion into joint entropies gives

$$\begin{aligned}\Delta I(X,Y) &= T_{Y\to X} - T^{\dagger}_{Y\to X} + T_{X\to Y} - T^{\dagger}_{X\to Y} + \alpha_2^{X,Y}, \\ \alpha_2^{X,Y} &:= I(X',Y'|X,Y) - I(X,Y|X',Y').\end{aligned} \tag{17}$$

Finally, the source-entropy representations

$$\begin{aligned}T_{Y\to X} &= H(Y) - I(X,Y) - H(Y|X,X'), \\ T^{\dagger}_{Y\to X} &= H(Y') - I(X',Y') - H(Y'|X,X')\end{aligned} \tag{18}$$

imply, upon taking a simultaneous time shift,

$$\begin{aligned}\Delta T_{Y\to X} &= \Delta H(Y) - \Delta I(X,Y) - \alpha_3^{Y\to X}, \\ \Delta T^{\dagger}_{Y\to X} &= \Delta H(Y) - \Delta I(X,Y) + \alpha_4^{Y} - \alpha_5^{X,Y} - \alpha_6^{Y\to X}.\end{aligned} \tag{19}$$

Here $\alpha_3^{Y\to X} = \Delta H(Y|X,X')$, $\alpha_4^{Y} = \Delta^2 H(Y)$, $\alpha_5^{X,Y} = \Delta^2 I(X,Y)$, and $\alpha_6^{Y\to X} = \Delta H(Y'|X,X')$. Substitution of the entropy and mutual-information balances gives the compact form used subsequently [30]:

$$\begin{cases}
\Delta H(\Gamma_i) = \displaystyle\sum_{k\in K_i} [T_{k\to\Gamma_i} - T^{\dagger}_{k\to\Gamma_i}] - E_{K_i\to\Gamma_i} + \alpha_1^{K_i\to\Gamma_i} \\
\Delta I(\Gamma_i,\Gamma_j) = [T_{\Gamma_j\to\Gamma_i} - T^{\dagger}_{\Gamma_j\to\Gamma_i}] \\
\qquad\qquad + [T_{\Gamma_i\to\Gamma_j} - T^{\dagger}_{\Gamma_i\to\Gamma_j}] + \alpha_2^{\Gamma_i,\Gamma_j} \\
\Delta T_{\Gamma_j\to\Gamma_i} = \displaystyle\sum_{k\in K_j\setminus\{\Gamma_i\}} [T_{k\to\Gamma_j} - T^{\dagger}_{k\to\Gamma_j}] \\
\qquad\qquad - [T_{\Gamma_j\to\Gamma_i} - T^{\dagger}_{\Gamma_j\to\Gamma_i}] - E_{K_j\to\Gamma_j} + \alpha_1^{K_j\to\Gamma_j} \\
\qquad\qquad - \alpha_2^{\Gamma_i,\Gamma_j} - \alpha_3^{\Gamma_j\to\Gamma_i} \\
\Delta T^{\dagger}_{\Gamma_j\to\Gamma_i} = \displaystyle\sum_{k\in K_j\setminus\{\Gamma_i\}} [T_{k\to\Gamma_j} - T^{\dagger}_{k\to\Gamma_j}] \\
\qquad\qquad - [T_{\Gamma_j\to\Gamma_i} - T^{\dagger}_{\Gamma_j\to\Gamma_i}] - E_{K_j\to\Gamma_j} + \alpha_1^{K_j\to\Gamma_j} \\
\qquad\qquad - \alpha_2^{\Gamma_i,\Gamma_j} + \alpha_4^{\Gamma_j} - \alpha_5^{\Gamma_i,\Gamma_j} - \alpha_6^{\Gamma_j\to\Gamma_i},
\end{cases} \tag{20}$$

with,

$$\begin{aligned}
\alpha_1^{K_i\to\Gamma_i} &:= H(\Gamma_i'|\Gamma_i, K_i) - H(\Gamma_i|\Gamma_i', K_i')\\
\alpha_2^{\Gamma_i,\Gamma_j} &:= I(\Gamma_i', \Gamma_j'|\Gamma_i, \Gamma_j) - I(\Gamma_i, \Gamma_j|\Gamma_i', \Gamma_j')\\
\alpha_3^{\Gamma_j\to\Gamma_i} &:= \Delta H(\Gamma_j|\Gamma_i, \Gamma_i')\\
\alpha_4^{\Gamma_j} &:= \Delta^2 H(\Gamma_j) = \Delta(\Delta H(\Gamma_j)) = H(\Gamma_j'') - 2H(\Gamma_j') + H(\Gamma_j)\\
\alpha_5^{\Gamma_i,\Gamma_j} &:= \Delta^2 I(\Gamma_i, \Gamma_j) = I(\Gamma_i'', \Gamma_j'') - 2I(\Gamma_i', \Gamma_j') + I(\Gamma_i, \Gamma_j)\\
\alpha_6^{\Gamma_j\to\Gamma_i} &:= \Delta H(\Gamma_j'|\Gamma_i, \Gamma_i').
\end{aligned} \tag{21}$$

In the last two equations we choose K_j to contain Γ_i; if necessary it can be enlarged, since the entropy identity is valid for any chosen collection. The α terms are residuals defined by the displayed entropies and time shifts. In particular, α_1 compares unresolved forward and backward uncertainty, α_2 compares conditional correlations across the two slices, α_3 and α_6 describe changes of conditional uncertainty, and α_4 and α_5 are second time differences.

The starting point is the stochastic process and its consistent joint laws, as specified in Sec. 2.1. The entropy chain rule then yields the displayed information balances. Once the process is fixed, both sides of every balance are determined. In particular, interpreting the terms as transfers and residuals reorganizes the evolution of the same probability laws; it does not introduce an additional source of randomness or an independent time parameter.

We retain the term *control variables* just for consistency with Ref. [30]. A *zero-control reduction* imposes specified vanishing residuals. Each subsequent proposition states which vanishing conditions are assumed, and whether they must hold at every step. Higher-order balance identities and their index sets are collected in Appendix B; the pairwise derivation above and Eq. (14) provide the balances needed to follow the main argument.

2.5. *Assumption of blocking flows*

The cycle calculations use a directional restriction on each interacting pair A, B:

$$\begin{cases} [T_{B\to A} = 0 \text{ and } T^\dagger_{A\to B} = 0] \\ \text{or } [T_{A\to B} = 0 \text{ and } T^\dagger_{B\to A} = 0]. \end{cases} \quad \text{at every time considered} \tag{22}$$

This is the *blocking-flows condition*. Its strong version additionally requires $T_{B\to A} > 0$ if and only if $T^\dagger_{A\to B} > 0$, for both orientations of each pair. Consequently, nonzero forward-transfer links have reversed counterparts in the reversed-transfer network. These are assumptions about the process; they do not follow from the definition of transfer entropy or from the presence of an update dependency. Section 3 uses the strong version, while the later calculations specify their own restrictions.

2.6. *Assumption of deterministic reversible dynamics*

This hypothesis is used for the cycle-necessity result of Sec. 3. The full-vector update is assumed to be deterministic and invertible at each step: $\exists\ U = (U_1, U_2, ..., U_N)$ and U^{-1} such that,

$$\begin{aligned} \Gamma'_i &= U_i(\Gamma_1, \ldots, \Gamma_N, t_0) \\ \Gamma_i &= (U^{-1})_i(\Gamma'_1, \ldots, \Gamma'_N, t_1), \end{aligned} \tag{23}$$

where $i \in \{1, 2, ..., N\}$ and $t_1 > t_0$. Then, with $K_i^{total} := \{\Gamma_1, \ldots, \Gamma_N\} - \{\Gamma_i\}$, the following uncertainty vanishes:

$$\begin{aligned} H(\Gamma'_i|\Gamma_i, K_i^{total}) &= 0 \\ H(\Gamma_i|\Gamma'_i, K'^{total}_i) &= 0. \end{aligned} \tag{24}$$

Hence, the control parameter $\alpha_1^{K_i^{total}\to\Gamma_i}$ defined in the equation (21) becomes zero in this case:

$$\alpha_1^{K_i^{total}\to\Gamma_i} := H(\Gamma'_i|\Gamma_i, K_i^{total}) - H(\Gamma_i|\Gamma'_i, K'^{total}_i) = 0. \tag{25}$$

Definition 6. A fixed collection $K_i = \{k_1, \ldots, k_m\} \subseteq \{\Gamma_j : j \neq i\}$ is a **sufficient forward/inverse dependency collection** for component i if its variables, together with the component's own state, suffice to write both local update relations over the interval considered:

$$\begin{aligned} \Gamma'_i &= U_i(\Gamma_i, k_1, \ldots, k_m; t), \\ \Gamma_i &= (U^{-1})_i(\Gamma'_i, k'_1, \ldots, k'_m; t+1). \end{aligned} \tag{26}$$

We use *set of dynamical neighbors* for such a collection in this specified structural sense. Forward dependency labels suffice for the first relation, inverse dependency labels suffice for the second, and their union is a sufficient collection. It is not required to be minimal and need not equal the direct forward-parent set of a time-unrolled graph.

Under the assumption of deterministic reversible dynamics, such a set exists, since the set K_i^{total} satisfies Eq. (26). Hence, by subtracting one by one from K_i^{total} while preserving Eq. (26), a smaller set of dynamical neighbors is obtained. Such a set need not be unique, and its elements need not all be direct forward parents: some may be needed only to reconstruct the inverse update. It is a sufficient two-direction dependency collection, rather than the parent set of a graph inferred from observations. For each component i, we choose one set K_i satisfying both relations in Eq. (26), and this choice is held fixed throughout Section 3.
As a result,

$$\alpha_1^{K_i\to\Gamma_i} := H(\Gamma'_i|\Gamma_i, K_i) - H(\Gamma_i|\Gamma'_i, K'_i) = 0. \tag{27}$$

2.7. *Assumption excluding purely synergistic incoming transfer*

Let the neighbor set of a component v be $K = \{k_1, \ldots, k_m\}$, and define an arbitrary but fixed tuple $\mathbf{K} := (k)_{k\in K}$. Then, the transfer entropy and reversed transfer entropy,

$$\begin{aligned} T_{\mathbf{K}\to v} &:= I(\mathbf{K}, v'|v) \\ T^{\dagger}_{\mathbf{K}\to v} &:= I(\mathbf{K}', v|v'), \end{aligned} \tag{28}$$

are distinct from the multiple transfer entropy $T_{K\to\{v\}}$ and multiple reversed transfer entropy $T^{\dagger}_{K\to\{v\}}$ defined in Sec. 2.3.

Decomposing $T_{\mathbf{K}\to v}$ and $T^{\dagger}_{\mathbf{K}\to v}$ yields:

$$\begin{aligned} T_{\mathbf{K}\to v} &= T_{k\to v} + I(\mathbf{K}_{\setminus k}, v'|v, k) \\ T^{\dagger}_{\mathbf{K}\to v} &= T^{\dagger}_{k\to v} + I(\mathbf{K}'_{\setminus k}, v|v', k'), \end{aligned} \tag{29}$$

for $k \in K$.

Since the right hand terms are non-negative, $T_{k\to v} = 0$ for all $k \in K$ if $T_{\mathbf{K}\to v} = 0$, and $T^{\dagger}_{k\to v} = 0$ for all $k \in K$ if $T^{\dagger}_{\mathbf{K}\to v} = 0$. Nevertheless, the converse is not generally true, as the counterexample is presented in Appendix A. To exclude incoming information that is invisible to every individual source, the analysis of Sec. 3 assumes that the converse holds, i.e., if $T_{k\to v} = 0$ for all $k \in K$, then $T_{\mathbf{K}\to v} = 0$, and if $T^{\dagger}_{k\to v} = 0$ for all $k \in K$, $T^{\dagger}_{\mathbf{K}\to v} = 0$. We refer to this condition as the non-synergy assumption for brevity. It is a zero-detection condition, not a claim that every co-information or correction E vanishes, or that all forms of synergy are absent. It excludes the pure synergistic effect, such as the XOR operation (Appendix A), while including, for example, the ternary additive receiver (Sec. 5.4.1).

3. General Information Dynamics of a Finite System

We now specify which graph is used and which hypotheses support the cycle result. A vertex labels a *component process* $\{\Gamma_i(t)\}_t$, not one state value and not a separate copy of a variable at one time. We take the vertex set to be the component labels $V = \{1, \ldots, N\}$ (or A, B, P, Q in the examples of Section 5). Expressions such as $H(v)$ and $T_{u\to v}$ abbreviate $H(\Gamma_v(t))$ and $I(\Gamma_v(t+1), \Gamma_u(t)|\Gamma_v(t))$. This component graph differs from a time-unrolled dependency graph, whose vertices would be (i, t).

The standing assumptions in this section are finite alphabets (Sec. 2.1), strong blocking flows (Sec. 2.5), deterministic reversible dynamics (Sec. 2.6), exclusion of purely synergistic incoming transfer in both forward and reversed directions (Sec. 2.7), fixed symmetric sufficient sets of dynamical neighbors K_v, and fixed nonempty information-flow support. The assumption excluding purely synergistic incoming transfer is imposed on these chosen sets. These assumptions are additional restrictions on a process, not generic properties of finite-state dynamics.

We use the following standard graph definitions [10, 27].

Definition 7. A **directed graph** is a pair $G = (V, \overrightarrow{E})$, where V is a finite set of vertices and $\overrightarrow{E} \subset \{(x, y) \in V \times V | x \neq y\}$ is a set of directed edges. Also, the reversed edge set $\overleftarrow{E} := \{(y, x) | (x, y) \in \overrightarrow{E}\}$ is defined. Then, the **in-degree** of a vertex $v \in V$ is $d^-(v) = |\{x \in V | (x, v) \in \overrightarrow{E}\}|$, while the **out-degree** of a vertex $v \in V$ is $d^+(v) = |\{x \in V | (v, x) \in \overrightarrow{E}\}|$.

Throughout this section, a directed graph $G = (V, \overrightarrow{E})$ is assumed to have no isolated vertices; that is, every vertex $v \in V$ satisfies $d^-(v) + d^+(v) > 0$.

Definition 8. A **source** in a directed graph $G = (V, \overrightarrow{E})$ is the vertex $v \in V$ with $d^-(v) = 0$ and $d^+(v) \neq 0$, and a **sink** in a directed graph $G = (V, \overrightarrow{E})$ is the vertex $v \in V$ with $d^+(v) = 0$ and $d^-(v) \neq 0$.

Definition 9. A **cycle** C in a directed graph $(V, \overrightarrow{E})$ is a sequence of distinct vertices $v_1, \ldots, v_n$, with $n \geqslant 2$, such that $(v_i, v_{i+1}) \in \overrightarrow{E}$ for $i = 1, \ldots, n-1$, and $(v_n, v_1) \in \overrightarrow{E}$.

Definition 10. A directed graph is **acyclic** if it contains no cycle.

Let $\mathcal{P}_v := \{u \in V \setminus \{v\} : \Gamma_u \in K_v\}$ be the label set corresponding to K_v. The chosen structural dependency graph has an edge $u \to v$ when $u \in \mathcal{P}_v$, with the sufficient-collection convention of Definition 6. Its edges record the selected structural neighborhood, including any dependencies needed for the inverse update. The graph used in the theorem is the *information-flow graph*

$$G_T(t) = (V, \overrightarrow{E}_T(t)), \qquad \overrightarrow{E}_T(t) := \{(u, v) : u \in \mathcal{P}_v,\ T_{u \to v}(t) > 0\}. \tag{30}$$

A structural dependency need not give a positive transfer entropy for a particular ensemble. An information edge records predictive dependence for the actual ensemble within the specified dependency neighborhood. *Fixed support* means that $\overrightarrow{E}_T(t)$ is the same edge set at every time in the interval considered; its positive weights may vary. Under the strong blocking condition and the symmetric neighborhood convention, the reversed-transfer graph has exactly the reversed edges. A cycle of G_T is what we call information circulation.

We discard isolated vertices and require at least one edge. Without this qualification, a collection of independent constant components would give stationary entropies and no information cycle. The result below concerns active information networks, and stationarity refers to constancy of the component-entropy vector.

Propositions 1 and 2 show that acyclic information networks necessarily exhibit monotonic entropy evolution, preventing stationary or periodic entropic dynamics under the present assumptions.

Proposition 1. *An acyclic graph must include at least one source and one sink.*

(proof) Start at a vertex and follow an incoming edge backward whenever possible. A vertex cannot be revisited, because that would produce a directed cycle. Finiteness therefore forces the path to stop at a vertex with no incoming edge. Since isolated vertices have been excluded, this vertex has an outgoing edge and is a source. Applying the same argument to the graph with every edge reversed gives a sink in the original graph. ■

While Proposition 1 is a classical result of graph theory, Proposition 2 focuses on the information dynamical consequences within the framework of Sec. 2.4, implying that, in acyclic information networks, individual entropy cannot be stationary or periodic.

Proposition 2. *Let $G_T = (V, \overrightarrow{E})$ be a directed network of pairwise transfer entropy with fixed support. Under the standing assumptions of this section and the non-synergistic assumption of Sec. 2.7, if G_T is acyclic, then the individual entropy of every sink is strictly increasing at each step of that interval.*

(proof) From Proposition 1, G_T has a sink $v \in V$: $\exists k \in K_v$ such that $T_{k\to v} > 0$, where K_v is the fixed set of dynamical neighbors of v. Then, the dynamics of v's individual entropy (Eq. (20)) is,

$$\begin{aligned}\Delta H(v) &= \sum_{k\in K_v} [T_{k\to v} - T^{\dagger}_{k\to v}] - E_{K_v\to v} + \alpha_1^{K_v\to v} \\ &= \sum_{k\in K_v} [T_{k\to v} - T^{\dagger}_{k\to v}] - E_{K_v\to v} \\ &= T_{\mathbf{K}_v\to v} - T^{\dagger}_{\mathbf{K}_v\to v}, \end{aligned} \tag{31}$$

where $\mathbf{K}_v := (k)_{k\in K_v}$.
Here, $\alpha_1^{K_v\to v} = 0$ due to the assumption of deterministic and reversible dynamics. Since v is a sink, $T_{v\to k} = 0$ for every $k \in K_v$. By the strong blocking flows condition, $T^{\dagger}_{k\to v} > 0$ would imply $T_{v\to k} > 0$, contradicting the sink property. Hence, $T^{\dagger}_{k\to v} = 0$ for every $k \in K_v$. The reversed non-synergy assumption of Section 2.7 requires $T^{\dagger}_{\mathbf{K}_v\to v} = 0$. In addition, by definition, $T_{\mathbf{K}_v\to v} := I(\mathbf{K}_v, v'|v) \geqslant 0$. If $T_{\mathbf{K}_v\to v} = 0$, then $T_{k\to v} = 0$ for all $k \in K_v$ by the argument in Section 2.7, which contradicts the fact that v is the sink.
Therefore,

$$\Delta H(v) = T_{\mathbf{K}_v\to v} > 0. \tag{32}$$

■

Since $\Delta H(v) > 0$ at every step, $H(v)$ is strictly increasing and therefore cannot be constant or periodic over any interval on which the support and the standing assumptions remain valid. If bounded, $H(v)$ may nevertheless converge monotonically to a limiting value. Consequently, one can conclude that the information-dynamical entropic state of a whole system, $\{H(u)\}_{u\in V}$, cannot be stationary or periodic in

this acyclic case, as the system has a node with such monotonic behavior. Therefore, we restrict our analysis to finite information networks with cycles to investigate other kinds of dynamical behaviors.

4. Information Dynamics of a Finite System with a Cycle

The information dynamics of finite systems containing a cycle is analyzed within the framework of Sec. 2, focusing on the behavior of a single cycle and its interaction with an external information source. Prior to that, the dynamics of an isolated cycle is considered.

4.1. *The Stationary State of a Single Cycle*

Let an N-component system $\Gamma = \{\Gamma_0, \Gamma_1, \ldots, \Gamma_{N-1}\}$ occupy the sites of a one-dimensional (1D) lattice with periodic boundary conditions. In the index notation, mod N is assumed in this section. Take $K_i = \{\Gamma_{i-1}, \Gamma_{i+1}\}$ and the oriented blocking conditions $T_{\Gamma_i \to \Gamma_{i-1}} = T^{\dagger}_{\Gamma_i \to \Gamma_{i+1}} = 0$. For this subsection, specialize the balances to $E_{K_i \to \Gamma_i} = \alpha_1^{K_i \to \Gamma_i} = \alpha_2^{\Gamma_i, \Gamma_{i+1}} = 0$ and $\alpha_3^{\Gamma_i \to \Gamma_{i+1}} = \alpha_6^{\Gamma_i \to \Gamma_{i-1}} = 0$, at every step. The zero residuals in the next time slice are included. Substituting into Eq. (20), or using the time-shifted identities in Appendix C, yields [30]

$$\begin{aligned}
\Delta H(\Gamma_i) &= T_{\Gamma_{i-1} \to \Gamma_i} - T^{\dagger}_{\Gamma_{i+1} \to \Gamma_i} \\
\Delta I(\Gamma_i, \Gamma_{i+1}) &= T_{\Gamma_i \to \Gamma_{i+1}} - T^{\dagger}_{\Gamma_{i+1} \to \Gamma_i} \\
\Delta T_{\Gamma_i \to \Gamma_{i+1}} &= T_{\Gamma_{i-1} \to \Gamma_i} - T_{\Gamma_i \to \Gamma_{i+1}} \\
\Delta T^{\dagger}_{\Gamma_i \to \Gamma_{i+1}} &= 0 \\
\Delta T_{\Gamma_i \to \Gamma_{i-1}} &= 0 \\
\Delta T^{\dagger}_{\Gamma_{i+1} \to \Gamma_i} &= T^{\dagger}_{\Gamma_i \to \Gamma_{i-1}} - T^{\dagger}_{\Gamma_{i+1} \to \Gamma_i}.
\end{aligned} \tag{33}$$

These equations describe the circulation of transfer entropy and reversed transfer entropy along the cycle. A stationary state is achieved when the transfer entropy and reversed transfer entropy are uniformly distributed along the cycle, satisfying $T_{\Gamma_{i-1} \to \Gamma_i} = T_{\Gamma_i \to \Gamma_{i+1}}$, $T^{\dagger}_{\Gamma_i \to \Gamma_{i-1}} = T^{\dagger}_{\Gamma_{i+1} \to \Gamma_i}$, and $T_{\Gamma_{i-1} \to \Gamma_i} = T^{\dagger}_{\Gamma_{i+1} \to \Gamma_i}$. This corresponds to a constant circulation of information along the cycle. As shown in Ref. [29], small deviations from this stationary state remain bounded in the examples analyzed there.

4.2. *The Transient Behaviors of the Single Cycle and a Random Source*

Set $t_1 = t_0 + 1$ and $t_2 = t_1 + 1$. Consider the interaction between a single cycle and an external information source. Let an external component Ext begin to interact with a component p in the cycle at the time $t = t_1$ as shown in Fig. 1, while no interaction is present at earlier times. Therefore, the stationary state of the previous section is

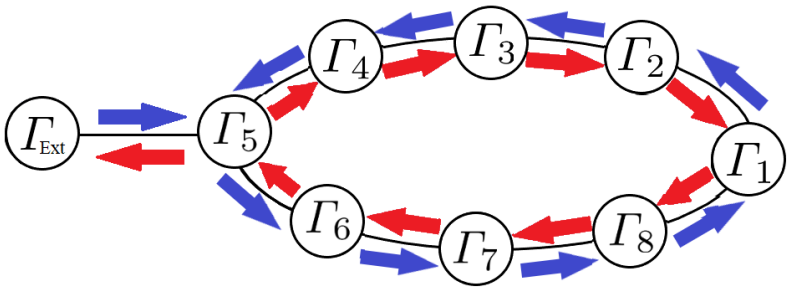


Fig. 1. Schematic illustrations of information dynamics on a cycle interacting with a random source (Ext). Blue arrows represent transfer entropy, and red arrows represent reversed transfer entropy.

assumed for the time $t = t_0 < t_1$. The interaction is a change in the update protocol: $T(t_0)$ uses the pre-activation transition $t_0 \to t_1$, whereas $T(t_1)$ uses $t_1 \to t_2$, the first activated transition. The coefficient abbreviation used below is

$$\alpha_{3456}^{Y\to X} := -\alpha_3^{Y\to X} - \alpha_4^{Y} + \alpha_5^{X,Y} + \alpha_6^{Y\to X}. \tag{34}$$

Proposition 3 gives the resulting balances under its explicitly stated residual conditions. It specializes the identities; it does not determine a response for an unspecified stochastic model.

Proposition 3. *Consider the activation protocol above with the blocking flows specified in Appendix C. Impose the residual conditions in Eq. (C.2) at every step, together with the initial condition* $E_{\{\Gamma_{i-1},\Gamma_{i+1}\}\to\Gamma_i} = 0$ *for* $i \neq p$. *The forward transfers adjacent to the receiving component obey*

$$\begin{aligned}
\Delta T_{\Gamma_{p-1}\to\Gamma_p} &= T_{\Gamma_{p-2}\to\Gamma_{p-1}} - T_{\Gamma_{p-1}\to\Gamma_p} - \alpha_3^{\Gamma_{p-1}\to\Gamma_p},\\
\Delta T_{\Gamma_{Ext}\to\Gamma_p} &= \alpha_1^{\{\Gamma_p\}\to\Gamma_{Ext}} - T_{\Gamma_{Ext}\to\Gamma_p} - \alpha_3^{\Gamma_{Ext}\to\Gamma_p},\\
\Delta T_{\Gamma_p\to\Gamma_{p+1}} &= T_{\Gamma_{p-1}\to\Gamma_p} + T_{\Gamma_{Ext}\to\Gamma_p} - T_{\Gamma_p\to\Gamma_{p+1}}\\
&\quad - E_{\{\Gamma_{p-1},\Gamma_{p+1},\Gamma_{Ext}\}\to\Gamma_p}.
\end{aligned} \tag{35}$$

The full system, including entropy, mutual information, and reversed-transfer balances, is given in Eq. (C.1) in Appendix C.

(proof) Appendix C. ■

These equations show that the external source modifies the information flow locally at component p, where the incoming flow $T_{\Gamma_{Ext}\to\Gamma_p}$ merges with the circu-

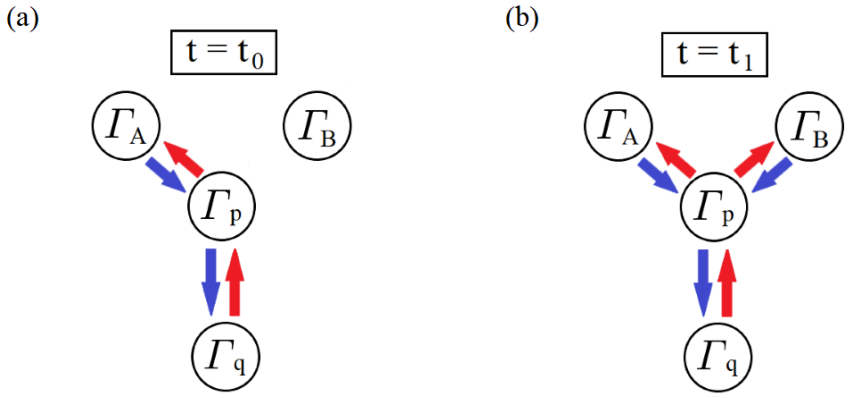


Fig. 2. Schematic illustration of the merging of two information flows at time **(a)** $t = t_0$ and **(b)** $t = t_1$. Blue arrows denote transfer entropy, and red arrows denote reversed transfer entropy.

lating flow $T_{\Gamma_{p-1}\to\Gamma_p}$. The corresponding control variables determine how the newly activated interaction is incorporated into the pre-existing circulation.

The changes at activation are recorded in the residual terms associated with the receiving component and the newly opened channel. For example, the external-transfer balance reads

$$\Delta T_{\Gamma_{Ext}\to\Gamma_p} = \alpha_1^{\{\Gamma_p\}\to\Gamma_{Ext}} - T_{\Gamma_{Ext}\to\Gamma_p} - \alpha_3^{\Gamma_{Ext}\to\Gamma_p}. \tag{36}$$

With zero baseline transfer, the first post-activation value is the difference of the two residual contributions on the right. Evaluating it requires the joint law generated by the source and receiver update rules. Thus the residuals describe the response; assigning them independently does not specify a realizable response.

Every actual stochastic process satisfies the exact balance identities, but arbitrary choices of the information variables and residuals need not correspond to consistent probability laws. In particular, a zero-control reduction can fail to respect the alphabet bound when extended beyond its range of applicability, as illustrated in Sec. 5.2. The next section derives an explicit capacity constraint directly from the receiver's conditional entropies.

5. The Merging of Two Information Flows

When a cycle of information flow interacts with an external information source, the manner in which two information flows merge at a single component determines the resulting dynamics. The merging of two information flows is analyzed within the framework of Sec. 2.4.

5.1. *Problem Setting*

Let A, B, P and Q be random variables defined over an ensemble, representing the states of four components. We consider a configuration in which the components represented by A and B interact with the component represented by P, while P also interacts with the component represented by Q (Fig. 2). At time $t = t_0$, the components A, P, and Q interact with each other, with the nonzero stationary transfer entropy $T_{A\to P} = T_{P\to Q} = \mu_A \neq 0$ and reversed transfer entropy $T^{\dagger}_{P\to A} = T^{\dagger}_{Q\to P} = \mu_A$, while the component B is not interacting with the component P, implying $T_{B\to P} = 0$ and $T^{\dagger}_{P\to B} = 0$. Then, the components B and P start to interact at the time $t = t_1$. Here $q_P := |\mathcal{M}_P|$ is fixed throughout, $C_P = \ln q_P$, and the downstream component Q is a separate random variable. The baseline assumptions include the stated blocking flows and zero higher-order correction before activation. The balance identities (Eq. (20)) express the first post-activation transfers as follows:

$$\begin{aligned}
T_{B'\to P'} &= \Delta T_{B\to P} = \sum_{k\in K_B\setminus\{P\}} [T_{k\to B} - T^{\dagger}_{k\to B}] \\
&\quad - E_{K_B\to B} + \alpha_1^{K_B\to B} - \alpha_2^{B,P} - \alpha_3^{B\to P}, \\
T^{\dagger}_{P'\to B'} &= \Delta T^{\dagger}_{P\to B} \\
&= [T_{A\to P} - T^{\dagger}_{Q\to P}] - E_{\{A,B,Q\}\to P} \\
&\quad + \alpha_1^{\{A,B,Q\}\to P} - \alpha_2^{B,P} + \alpha_4^{P} - \alpha_5^{B,P} - \alpha_6^{P\to B}.
\end{aligned} \tag{37}$$

The conditions $T_{B\to P} = T^{\dagger}_{P\to B} = 0$ and the stationary conditions $\Delta I(B, P) = 0$ imply that $\alpha_2^{B,P} = 0$. We also assume the pre-activation baseline $E_{\{A,B,Q\}\to P} = E_{\{A,Q\}\to P} = 0$. This is a condition on the baseline joint law; absence of a pairwise transfer from B alone would not establish it, because higher-order dependence can be hidden from pairwise measures.

$$\begin{aligned}
T_{B'\to P'} &= \sum_{k\in K_B\setminus\{P\}} [T_{k\to B} - T^{\dagger}_{k\to B}] - E_{K_B\to B} \\
&\quad + \alpha_1^{K_B\to B} - \alpha_3^{B\to P}, \\
T^{\dagger}_{P'\to B'} &= [T_{A\to P} - T^{\dagger}_{Q\to P}] + \alpha_1^{\{A,B,Q\}\to P} \\
&\quad + \alpha_4^{P} - \alpha_5^{B,P} - \alpha_6^{P\to B}.
\end{aligned} \tag{38}$$

5.2. *The Zero-control Limit*

The structural role of control variables α in the merge problem is recognized by analyzing the uncontrolled behavior of dynamical variables. In the zero-control limit with the blocking flow condition, all control variables vanish except those associated

with blocked channels. Then, the equation (38) in the limit becomes,

$$T_{B'\to P'} = \sum_{k\in K_B\setminus\{P\}} [T_{k\to B} - T^{\dagger}_{k\to B}] - E_{K_B\to B}. \tag{39}$$

In this reduction, $T_{B'\to P'}$ is expressed in terms of information flows into B and the correction $E_{K_B\to B}$. Since $T_{A'\to P'} = T_{A\to P} = \mu_A$ and $T^{\dagger}_{Q'\to P'} = T^{\dagger}_{Q\to P} = \mu_A$ by the zero-control limit,

$$\Delta H(P') = [T_{A'\to P'} + T_{B'\to P'} - T^{\dagger}_{Q'\to P'}] = T_{B'\to P'}. \tag{40}$$

Consequently, the formal zero-control evolution does not itself prevent continued increase of $H(P')$. However, such unrestricted growth is not generally realizable. The entropy of component P must satisfy $H(P') \leqslant C_P$, with $C_P = \ln 2$ for a binary state space. The reduced balance does not explicitly enforce the receiver's entropy-capacity constraint. The finite maximum is already guaranteed by the alphabet. Rather, a realizable process must have residual terms consistent with that bound; setting all relevant residuals to zero need not remain compatible with the assumed transfers. The residuals record this constraint and do not create the entropy ceiling.

5.3. *Receiver assumptions and the remaining conditional-entropy margin*

Define the remaining conditional-entropy margin at the pre-activation transition by

$$\epsilon := C_P - H(P'|P) \geq 0. \tag{41}$$

Thus $\epsilon \to 0$ means saturation of a *conditional* entropy ceiling, a stronger statement than $H(P') \to C_P$ alone. The receiver result uses the following six additional assumptions. Any derivative is taken along a differentiable family of consistent joint laws on the same fixed alphabet, parameterized locally by ϵ; it is not a derivative in physical time. Below we show that this operational definition agrees with Eq. (46).

$$\begin{aligned}
&\text{Assumption 1: } T_{A'\to P'} - T_{\{A',B'\}\to P'} \\
&\qquad\qquad = I(A', P''|P', B') = T_{A\to P} \\
&\text{Assumption 2 : } I(Q, P'|P, A) = 0,\ I(Q', P''|P', A', B') = 0 \\
&\text{Assumption 3 : } \alpha_1^{\{A,Q\}\to P} = \alpha_1^{\{A',B',Q'\}\to P'} \\
&\text{Assumption 4 : } \frac{\partial \alpha_1^{\{A,Q\}\to P}}{\partial \epsilon} > -1 \\
&\text{Assumption 5 : } Q' = P, Q'' = P' \\
&\text{Assumption 6 : } \Delta I(P', P) + T_{B'\to P'} \geqslant 0,
\end{aligned} \tag{42}$$

where ϵ is defined in Eq. (41).

Assumption 1 preserves the baseline contribution from A after conditioning on the newly active input B'. It does not assert that the signed co-information term vanishes. Assumption 2 says that the downstream variable provides no additional predictive information once the listed receiver and source states are known.

Assumption 3 equates the residual unresolved uncertainty before and after activation. Assumption 4 specifies how that residual varies along the chosen family; it is needed to relate increased baseline transfer to a reduced margin. Assumption 5 makes Q a one-step delay copy of P. Assumption 6 is a sign condition on the change of the receiver's lag-one mutual information and the external transfer, where $\Delta I(P', P) = I(P'', P') - I(P', P)$. These conditions define the class analyzed here; finite capacity alone does not imply them.

From Eq. (20),

$$\begin{aligned} T_{B'\to P'} =& \Delta H(P') + T^{\dagger}_{Q'\to P'} - T_{A'\to P'} \\ &+ E_{\{A',B',Q'\}\to P'} - \alpha_1^{\{A',B',Q'\}\to P'}. \end{aligned} \tag{43}$$

The term $E_{\{A',B',Q'\}\to P'}$ can be reduced to the single multiple transfer entropy:

Proposition 4. *From Assumption 2, 5,*

$$E_{\{A',B',Q'\}\to P'} = T_{\{A',B'\}\to P'}. \tag{44}$$

(proof) Appendix D. ■

Proposition 4 and Assumption 1 lead to:

$$T_{B'\to P'} = \Delta H(P') + T^{\dagger}_{Q'\to P'} - T_{A\to P} - \alpha_1^{\{A',B',Q'\}\to P'}. \tag{45}$$

The same margin ϵ can be written as

$$\epsilon := C_P - \alpha_1^{\{A,Q\}\to P} - T_{A\to P}. \tag{46}$$

Under Assumptions 2 and 5, the chain rule gives $H(P'|P) = T_{A\to P} + \alpha_1^{\{A,Q\}\to P}$, as shown in Appendix E. Here $\alpha_1 = H(P'|P, A) \geq 0$ is the unresolved forward uncertainty in this receiver construction; this interpretation does not apply to an arbitrary signed α_1. Thus Eq. (46) is precisely the margin of Eq. (41). Additionally, define δ by:

$$\delta := C_P - \Delta H(P') - T^{\dagger}_{Q'\to P'}, \tag{47}$$

indicating,

$$T_{B'\to P'} = \epsilon - \delta, \tag{48}$$

from Eq. (45) and Assumption 3.

Before discussing the relationship between ϵ and δ, one can deduce the lemma.

Lemma 1. *Under Assumptions 1, 2, 3, and 5,*

$$H(P''|P') = H(P'|P) + T_{B'\to P'}. \tag{49}$$

(proof) Appendix E. ■

Appendix E also establishes that Assumptions 2 and 5 imply $H(P'|P) = \alpha_1^{\{A,Q\}\to P} + T_{A\to P}$. Consequently, the limit $\epsilon \to 0$ corresponds to the saturation of the conditional-entropy ceiling: $\epsilon = C_P - H(P'|P)$.

The main capacity bound now has a short proof. The lemma and the alphabet ceiling give

$$0 \leq T_{B' \to P'} = H(P''|P') - H(P'|P) \leq C_P - H(P'|P) = \epsilon. \tag{50}$$

■

Assumptions 1, 2, 3, and 5 suffice for this identity and bound. Assumption 4 is used to interpret how the margin varies with the baseline transfer, while Assumption 6 additionally controls the sign of the change in marginal entropy. We next record the auxiliary bounds needed for the boundary analysis.

Proposition 5. *From Assumption 5, 6 and Lemma 1,*

$$\begin{aligned} &(a)\ H(P') \geqslant C_P - \epsilon \\ &(b)\ 0 \leqslant \delta \leqslant \epsilon \\ &(c)\ 0 \leqslant \Delta H(P') \leqslant \epsilon. \end{aligned} \tag{51}$$

(proof) Appendix F. ■

From Proposition 5, one obtains,

$$\begin{aligned} 0 \leqslant T_{B' \to P'} \leqslant \epsilon &= C_P - \alpha_1^{\{A,Q\} \to P} - T_{A \to P} \\ &= C_P - H(P'|P). \end{aligned} \tag{52}$$

Equation (52) bounds the admissible external transfer entropy by ϵ, the remaining conditional-entropy margin. It therefore establishes competition at the level of an upper bound. Under Assumption 4, increasing the pre-existing transfer entropy reduces ϵ and hence lowers the admissible upper bound on the external transfer entropy. A corresponding decrease in the realized transfer entropy does not follow from the bound alone and must be verified for each specified model. The control variables α_3 and α_6 must accordingly satisfy the conditions leading to Eq. (52) as the conditional-entropy margin approaches zero.

In communication theory [11, 37], this competition is analogous to cross-talk in a two-sender-one-receiver multiple-access channel (MAC). The cross-talk [20, 33, 58] in MAC has been referred to as the difficulty of simultaneous communication with two senders, when the receiver's capacity for information processing is limited. Hence, Eq. (52) states that, as the pre-existing transfer from A to P increases, the admissible upper bound on transfer from B decreases when $H(P'|P)$ is close to C_P. A multiplex receiver can avoid this competition by assigning separate state-space coordinates to the two inputs, as illustrated in Sec. 5.4.2.

Intuitively, a finite-capacity receiving component cannot accommodate arbitrarily many independent incoming influences. Eq. (52) makes this competition quantitative by bounding the additional transfer entropy in terms of the remaining conditional-entropy margin, expressing suppression of admissible external information transfer.

Furthermore, under limiting conditions, we derive a more tractable formulation, since the upper bound of Eq. (52) differs from the realized transfer entropy $T_{B' \to P'}$.

The defining saturation condition is $H(P'|P) \to C_P$. In this limit, Proposition 5 additionally implies,

$$\begin{aligned} &H(P') \to C_P \\ &\Delta H(P') \to 0 \\ &T^{\dagger}_{Q' \to P'} \to C_P \\ &T_{B' \to P'} \to 0 \\ &H(P''|P') \to C_P. \end{aligned} \tag{53}$$

From this limit, $T_{B' \to P'}$ can be expanded as:

$$T_{B' \to P'}(\epsilon) = \frac{dT_{B' \to P'}}{d\epsilon}|_{\epsilon=0^+}\epsilon + o(\epsilon), \tag{54}$$

assuming the differentiability of $T_{B' \to P'}$ near $\epsilon = 0$.
By Eq. (52),

$$0 \leqslant \left.\frac{dT_{B' \to P'}}{d\epsilon}\right|_{\epsilon=0^+} \leqslant 1, \tag{55}$$

indicating that the derivative at $\epsilon = 0$ is non-negative. Moreover, Assumption 4 ensures from Eq. (46) that $dT_{A \to P}/d\epsilon < 0$; namely, increasing the pre-existing flow reduces the remaining entropy-capacity margin. Hence, under this one-sided boundary condition, the derivative ratio can be expressed as:

$$\begin{aligned} \left.\frac{dT_{B' \to P'}}{dT_{A \to P}}\right|_{\epsilon=0^+} &= \frac{\left.\dfrac{dT_{B' \to P'}}{d\epsilon}\right|_{\epsilon=0^+}}{\left.\dfrac{dT_{A \to P}}{d\epsilon}\right|_{\epsilon=0^+}} \\ &\leqslant 0. \end{aligned} \tag{56}$$

Up to this point, ϵ is defined by Eq. (46), while $T_{B' \to P'}$ has been expanded based on its limiting behavior. No finite-margin monotonicity follows from the boundary result alone. To examine finite-margin behavior, we consider a parameterized family $\{p_\theta\}_{\theta \in \Theta}$ of underlying probability distributions on a fixed alphabet. In this case, $T_{A \to P}$ and $T_{B' \to P'}$ can be denoted $T_{A \to P}(\theta)$ and $T_{B' \to P'}(\theta)$, respectively. For such a family, we test whether the following additional model-dependent condition holds over a finite range of θ:

$$\begin{aligned} \frac{dT_{B' \to P'}}{dT_{A \to P}}(\theta) &= \frac{dT_{B' \to P'}(\theta)/d\theta}{dT_{A \to P}(\theta)/d\theta} \\ &\leqslant 0. \end{aligned} \tag{57}$$

Eq. (57) characterizes local suppression of the newly incoming flow with $\frac{dT_{A \to P}}{d\theta} \neq 0$. In Section 5.4, we examine Eq. (57) for explicit statistical models.

5.4. *Examples of merging information flows*

The following examples examine separately whether the hypotheses of the boundary result (Eq. (42)) are satisfied and whether a particular model exhibits a negative relation, Eq. (57), over a finite parameter range.

5.4.1. *Ternary Additive Receiver*

Let the dynamics at time $t < t_1$ be,

$$\begin{aligned} A' &\sim p_a \\ B' &\sim p_b \\ P' &= A \\ Q' &= P, \qquad \text{for } \ t < t_1 \end{aligned} \tag{58}$$

where $p_\theta := (1-2\theta, \theta, \theta)$ is a one-parameter distribution on $\mathbb{Z}_3$ with $0 < \theta < \frac{1}{3}$ and A' and B' belong to mutually independent i.i.d. time sequences distributed according to p_a and p_b, respectively.

At time $t \geqslant t_1$,

$$\begin{aligned} A'' &\sim p_a \\ B'' &\sim p_b \\ P'' &= A' + B' \qquad \mod 3 \\ Q'' &= P', \end{aligned} \tag{59}$$

Then,

$$T_{A\to P} = H(A) = h(a), \tag{60}$$

where $h(x) := -(1-2x)\ln(1-2x) - (2x)\ln(x)$.
Therefore,

$$\frac{dT_{A\to P}}{da} = h'(a) = 2\ \ln\frac{1-2a}{a} > 0. \tag{61}$$

Since $\mathbb{P}(P''=0) = 1-2(a+b-3ab)$ and $\mathbb{P}(P''=1) = \mathbb{P}(P''=2) = a+b-3ab$,

$$\begin{aligned} T_{B'\to P'} &= I(P'', B') = H(P'') - H(A') \\ &= h(a+b-3ab) - h(a). \end{aligned} \tag{62}$$

This yields,

$$\begin{aligned} \frac{dT_{B'\to P'}}{da} &= (1-3b)h'(a+b-3ab) - h'(a) < 0, \\ &\text{because } h''(x) < 0 \text{ for } 0 < x < \tfrac{1}{3}, \\ &\text{and } 0 < 1-3b < 1. \end{aligned} \tag{63}$$

Hence,

$$\frac{dT_{B'\to P'}}{dT_{A\to P}} = \frac{\frac{dT_{B'\to P'}}{da}}{\frac{dT_{A\to P}}{da}} < 0. \tag{64}$$

For this dynamics, the equation (57) is valid for the overall range. Also, it can be shown that the assumptions of Eq. (42) are valid.

5.4.2. *A Multiplex Receiver*

On the other hand, most artificial networks, such as multi-source networks in communication theory, are designed to prevent the cross-talk effect for a reliable communication. This is accomplished by modeling the multiplex receiver to process input signals independently. Therefore, the multiplex receiver is assigned two independent state-space components, $P = (P_1, P_2)$. Then, the dynamics is set as,

$$\begin{aligned} A' &\sim Uniform(N_{Z_A}) \\ B' &\sim Uniform(N_{Z_B}) \\ P_1' &= A \\ P_2' &= P_2 \\ Q_1' &= P_1 \\ Q_2' &= P_2, \qquad\qquad \text{for } \; t < t_1 \end{aligned} \tag{65}$$

where $N_Z := \{0, 1, \ldots, Z-1\}$ and Z_A, $Z_B \in \mathbb{N}$, and $Uniform(N_Z)$ is the uniform distribution for the set N_Z.

$$\begin{aligned} A'' &\sim Uniform(N_{Z_A}) \\ B'' &\sim Uniform(N_{Z_B}) \\ P_1'' &= A' \\ P_2'' &= B' \\ Q_1'' &= P_1' \\ Q_2'' &= P_2'. \qquad\qquad \text{for } \; t \geqslant t_1 \end{aligned} \tag{66}$$

As a result,

$$\begin{aligned} T_{A\to P} &= \ln(Z_A) \\ T_{B'\to P'} &= \ln(Z_B), \end{aligned} \tag{67}$$

and, thus, $T_{B'\to P'}$ is independent of Z_A and hence of $T_{A\to P}$ with a fixed Z_B. Since this construction avoids competition by supplying an additional receiver coordinate, it is a contrast model rather than a fixed-alphabet test of the boundary derivative. Consequently, this represents a limiting case in which the cross-talk effect is eliminated.

5.4.3. *AND Gate*

Unlike the previous examples, the merge with AND gate violates some of the assumptions and does not satisfy Eq. (57) in general.

$$\begin{aligned} A' &\sim B(\frac{1}{2}+\rho_A) \\ B' &\sim B(\frac{1}{2}+\rho_B) \\ P' &= A \\ Q' &= P, \qquad\qquad \text{for } \ t < t_1 \end{aligned} \tag{68}$$

where $B(\rho)$ is the Bernoulli distribution with success probability ρ and $-\frac{1}{2} \leqslant \rho_A, \rho_B \leqslant \frac{1}{2}$.

$$\begin{aligned} A'' &\sim B(\frac{1}{2}+\rho_A) \\ B'' &\sim B(\frac{1}{2}+\rho_B) \\ P'' &= AND(A', B') \\ Q'' &= P'. \qquad\qquad \text{for } \ t \geqslant t_1 \end{aligned} \tag{69}$$

Here,

$$\begin{aligned} T_{A\to P} = & -(\tfrac{1}{2}+\rho_A)\ln(\tfrac{1}{2}+\rho_A) \\ & -(\tfrac{1}{2}-\rho_A)\ln(\tfrac{1}{2}-\rho_A), \\ T_{B'\to P'} = & -(\tfrac{1}{2}+\rho_A)(\tfrac{1}{2}+\rho_B)\ln(\tfrac{1}{2}+\rho_B) \\ & +(\tfrac{1}{2}-\rho_A)(\tfrac{1}{2}+\rho_B)\ln(\tfrac{1}{2}-\rho_A) \\ & -\big[1-(\tfrac{1}{2}+\rho_A)(\tfrac{1}{2}+\rho_B)\big] \\ & \quad\times\ln\big[1-(\tfrac{1}{2}+\rho_A)(\tfrac{1}{2}+\rho_B)\big]. \end{aligned} \tag{70}$$

While,

$$\begin{aligned} \frac{\partial T_{B'\to P'}}{\partial \rho_A} &= (\tfrac{1}{2}+\rho_B) \\ &\qquad \times \ln\frac{1-(\frac{1}{2}+\rho_A)(\frac{1}{2}+\rho_B)}{(\frac{1}{2}-\rho_A)(\frac{1}{2}+\rho_B)} \geqslant 0, \end{aligned} \tag{71}$$

$$\frac{\partial T_{A\to P}}{\partial \rho_A} = \ln\frac{\frac{1}{2}-\rho_A}{\frac{1}{2}+\rho_A} \begin{cases} <0 & (\rho_A > 0) \\ >0 & (\rho_A < 0) \\ =0 & (\rho_A = 0) \end{cases} \tag{72}$$

Therefore,

$$\frac{dT_{B'\to P'}}{dT_{A\to P}} \begin{cases} \leqslant 0 & (\rho_A > 0) \\ \geqslant 0 & (\rho_A < 0) \end{cases} \tag{73}$$

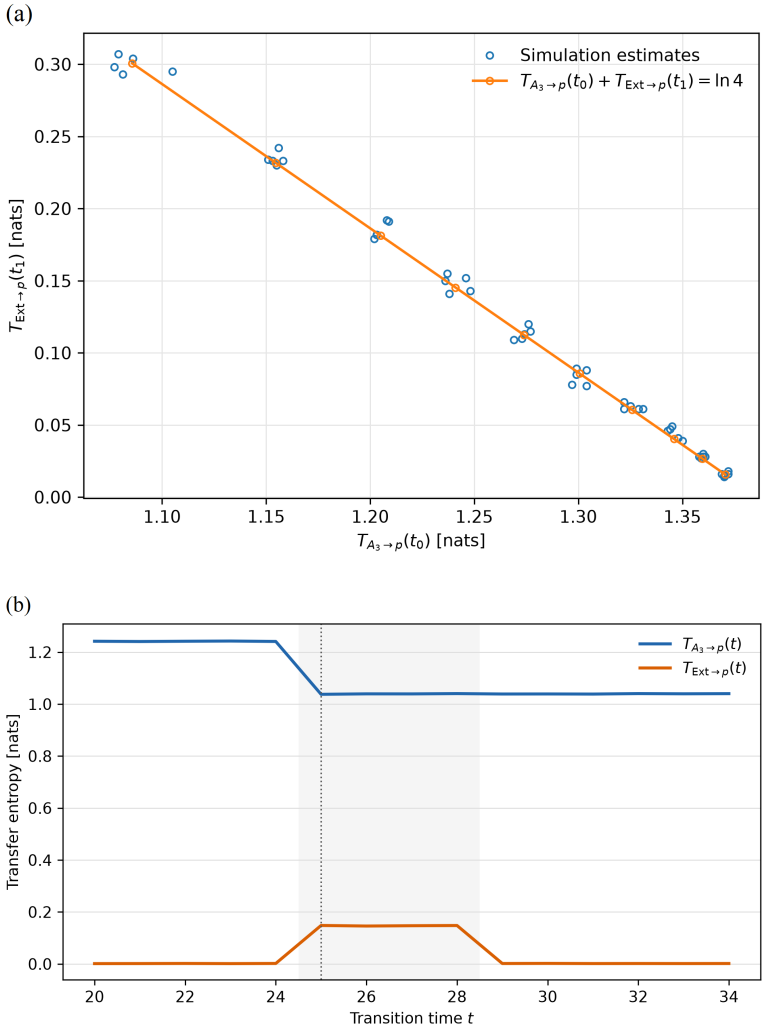


Fig. 3. **(a)** Simulation results for competing information flows: $T_{A3\to P}(t_0)$ versus $T_{Ext\to P}(t_1)$. The cycle $P \to A1 \to \cdots \to A3 \to P$ is isolated at time $t = t_0$, and node P begins to interact with the external source (Ext) at $t = t_1$. **(b)** The time-evolution of $T_{A3\to P}$ and $T_{Ext\to P}$ with $\theta = 0.18$. Here, $t_1 = 25$ time steps.

Assumptions 2, 3, and 5 are satisfied; however, Assumption 1 fails except the trivial case where $\rho_B = \frac{1}{2}$, and the validity of Assumption 4 depends on the value of ρ_A and ρ_B.

6. A two-bit circulation with an activated external input

To illustrate the mechanism derived in Sec. 5, we construct a stochastic network with the Fredkin gate [15], which is a standard gate in the literature of reversible computing [6,34,39,51]: a cycle $P \to A1 \to \cdots \to A3 \to P$ and an external node (Ext). Each node has a two-bit state, $\Gamma_{A1}, \ldots, \Gamma_{A3}, \Gamma_P, \Gamma_{Ext} \in \{(0,0),(0,1),(1,0),(1,1)\}$. The receiver capacity is $C_P = \ln 4$, and the update rule at the merging node P is given by a Fredkin-type controlled-SWAP gate:

$$\Gamma'_P = \begin{cases} (\gamma^{(2)}_{A3}, \gamma^{(1)}_{A3}) & \text{if } \gamma^{(1)}_{Ext} = 1 \\ (\gamma^{(1)}_{A3}, \gamma^{(2)}_{A3}) & \text{if } \gamma^{(1)}_{Ext} = 0 \end{cases}, \tag{74}$$

where $\Gamma_X = (\gamma^{(1)}_X, \gamma^{(2)}_X)$ for a node X with $\gamma^{(1)}_X, \gamma^{(2)}_X \in \{0,1\}$.

Although the gate originates from conservative logic, we analyze its information dynamics under a parameterized input ensemble. For $0 \leqslant \theta < 1/4$, we define the probability mass function p_θ such that, for a random variable $\Gamma_X \sim p_\theta$,

$$\begin{aligned} p_\theta(\Gamma_X = (1,0)) &= \frac{1}{4} + \theta \\ p_\theta(\Gamma_X = (0,1)) &= \frac{1}{4} - \theta \\ p_\theta(\Gamma_X = (0,0)) &= p_\theta(\Gamma_X = (1,1)) = \frac{1}{4}. \end{aligned} \tag{75}$$

Then, the initial ensemble at $t = t_0$ is constructed as:

$$\begin{aligned} &\Gamma_{A1}, \Gamma_{A2}, \Gamma_{A3} \sim p_\theta \\ &\Gamma_P \sim p_\theta \\ &\Gamma_{Ext} \sim p_0, \end{aligned} \tag{76}$$

where $\Gamma_{A1}, \ldots, \Gamma_{A3}, \Gamma_P, \Gamma_{Ext}$ are mutually independent. The source $\Gamma_{Ext}(t)$ is independently resampled from p_0 at every step, independently of the initial cycle state. Before activation, $\Gamma'_P = \Gamma_{A3}$; after activation, the controlled-SWAP rule above is used. The remaining updates are $\Gamma'_{A1} = \Gamma_P$, $\Gamma'_{A2} = \Gamma_{A1}$, and $\Gamma'_{A3} = \Gamma_{A2}$. The update $\Gamma'_{A1} = \Gamma_P$ provides the delay-copy relation of Assumption 5 in Sec. 5. The full system with a resampled source is stochastic; reversibility of the local gate does not assert global deterministic reversibility of this example.

The resulting dynamics satisfies all assumptions of Eq. (42): for the construction in this section, Assumptions 1, 2, and 5 follow directly from the conditional bijectivity of the controlled-SWAP operation and the delay-copy relation. At the onset of interaction, independence of the node states gives $\alpha_1^{\{A,Q\}\to P} = \alpha_1^{\{A',B',Q'\}\to P'} = 0$, $\frac{\partial \alpha_1^{\{A,Q\}\to P}}{\partial \epsilon} = 0$, $\Delta I(P', P) = 0$, so Assumptions 3, 4, and 6 are also satisfied. Then, the exact calculation yields:

$$T_{A3\to P}(t_0) + T_{Ext\to P}(t_1) = C_P, \tag{77}$$

where $T_{A3\to P}(t_0) := T_{A3\to P}$ and $T_{Ext\to P}(t_1) := T_{Ext'\to P'}$.
At the onset of interaction, Eq. (77) shows that the realized external transfer en-

tropy equals the remaining capacity margin. Consequently, within this parameterized Fredkin model, increasing $T_{A3\to P}(t_0)$ is accompanied by a corresponding decrease in $T_{Ext\to P}(t_1)$, thereby realizing the finite-range suppression relation in Eq. (57).

Figure 3(a) compares different prepared ensembles indexed by θ, at the two specified transitions. Its horizontal axis is a baseline transfer varied across ensembles, rather than elapsed time. To examine physical-time evolution, Fig. 3(b) instead fixes θ. We simulate an ensemble of independent realizations, collect state pairs at the same two times across realizations, and use empirical frequency tables on the finite joint alphabet to estimate the information measures. No stationarity over the transient is assumed. Simulation results (Fig. 3 (a)) [29] show that increasing the internal circulating flow $T_{A3\to P}(t_0)$ suppresses the external input $T_{Ext\to P}(t_1)$ in accordance with Eq. (77), quantitatively confirming the relation (Eq. (57)).
Furthermore, Fig. 3 (b) shows that both $T_{A3\to P}(t)$ and $T_{Ext\to P}(t)$ evolve over time. Following activation of the interaction, $T_{Ext\to P}(t)$ initially increases to a plateau before dissipating to zero as the probability distribution mixed by (Ext) propagates through the cycle to node P. Since the swapping operation by (Ext) symmetrizes the asymmetry induced by parameter θ in the initial ensemble, the node P ceases to receive influence from node (Ext) under this symmetrized distribution. This demonstrates that the competition between information flows exhibits distinct temporal characteristics.

7. Conclusion

This work establishes two information-dynamical results. First, the circulation of transfer entropy and reversed transfer entropy constitutes a necessary structural condition for stationary or periodic entropic dynamics in active finite-state networks under the stated assumptions and fixed nonempty support. Second, when a preexisting circulating flow and an external flow merge at a finite-capacity node, the remaining conditional-entropy margin provides an upper bound on the admissible external transfer entropy. In this sense, the bound identifies a possible information-transfer constraint associated with cyclic organization. This mechanism corresponds to a cross-talk effect, in which limited entropy capacity constrains simultaneous information transfer.

For the Fredkin model considered here, the bound is attained at the onset of interaction, producing a model-specific negative relation between the circulating and external transfer entropies. The subsequent simulation shows that this relation is transient and need not remain saturated after the modified distribution propagates around the cycle. Consequently, the quantitative framework developed in this work facilitates an information-dynamical analysis of reversible logic gates.

The current analysis relies on several assumptions introduced in Sec. 2 and Eq. (42). While these assumptions facilitate tractable analysis of interacting information flows and yield baseline cases, their necessity and generality remain to be examined.

Further work is required to relax these assumptions, explore a wider class of merging mechanisms, and investigate additional properties of information circulation, including its temporal structure and interactions between multiple cycles.

A further limitation concerns the admissibility of the information-dynamical evolution, as discussed in Section 4.2 and Ref. [30]. A complete characterization of the admissible region of the control variables, subject to Shannon inequalities, alphabet bounds, and temporal consistency of the underlying distributions, remains an open problem.

Furthermore, the present analysis assumes that the directed structure of transfer entropy interactions remains sufficiently stable over the analyzed time interval. Extending the framework to adaptive networks, in which the interaction topology and effective information flow structure evolve dynamically, remains an important direction for future research.

Data Availability Statement

The data that support the findings of this study are openly available in Zenodo at https://doi.org/10.5281/zenodo.22358615 (Version v0.1.4b).

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Appendix A. A Counterexample to the Assumption of Section 2.7

Let the random variables u_1, u_2, v represent the state of three components in discrete time having the dynamics:

$$\begin{aligned} u_1', u_2' &\sim B(\frac{1}{2}) \\ v' &= u_1 \oplus u_2, \end{aligned} \tag{A.1}$$

where $B(p)$ is the Bernoulli distribution with success probability p, and the operation $\oplus$ represents the binary operation XOR.
Then, $H(v'|v) = H(v'|v, u_1) = H(v'|v, u_2) = \ln 2$, resulting in $T_{u_1 \to v} = T_{u_2 \to v} = 0$. Nonetheless, for $\mathbf{u} = (u_1, u_2)$, $T_{\mathbf{u} \to v} = H(v'|v) - H(v'|v, u_1, u_2) = \ln 2$.

Appendix B. Higher-order information balances

This appendix records the multivariate hierarchy used for the higher-order cancellations in Appendix C. All variables and time shifts follow Sec. 2. Here $I(V)$ and $I(V|W)$ are co-informations, including the singleton convention $I(\{X\}) = H(X)$. Transfer quantities with collection subscripts are the signed terms of Eq. (11).

For a collection V containing at least two variables, define the set of ordered nonempty bipartitions by

$$D_V := \{(U, W) : U \cap W = \varnothing,\ U \cup W = V,\ U \neq \varnothing,\ W \neq \varnothing\}. \tag{B.1}$$

Thus (U, W) and (W, U) are both included. All subset sums below range over collections of distinct labeled variables. These identities follow by applying inclusion–exclusion to the same entropy chain rules as the pairwise balances [30]:

$$\begin{cases} E_{K_i \to \Gamma_i} := \sum_{u=2}^{|K_i|} (-1)^u \sum_{\substack{V \subset K_i \\ |V|=u}} [T_{V \to \Gamma_i} - T^{\dagger}_{V \to \Gamma_i}] \\ \Delta I(V) = \sum_{(U,W) \in D_V} [T_{U \to W} - T^{\dagger}_{U \to W}] + \alpha_2^V \\ \Delta T_{U \to W} = (-1)^{|W|} \Delta I(U \cup W) \\ \qquad + \sum_{u=1}^{|W|} (-1)^{|W|-u} \sum_{\substack{S \subset W \\ |S|=u}} \Big[\Delta I\big(U \cup (W \setminus S)\big) \\ \qquad\quad + \sum_{s=1}^{u} (-1)^s \sum_{\substack{Z \subset S \\ |Z|=s}} \alpha_3^{U \cup (W \setminus S) \to Z} \Big] \\ \Delta T^{\dagger}_{U \to W} = (-1)^{|W|} \Delta I(U \cup W) + (-1)^{|W|} \alpha_5^{U \cup W} \\ \qquad + \sum_{u=1}^{|W|} (-1)^{|W|-u} \sum_{\substack{S \subset W \\ |S|=u}} \Big[\Delta I\big(U \cup (W \setminus S)\big) \\ \qquad\quad + \alpha_5^{U \cup (W \setminus S)} + \sum_{s=1}^{u} (-1)^s \sum_{\substack{Z \subset S \\ |Z|=s}} \alpha_6^{U \cup (W \setminus S) \to Z} \Big], \end{cases} \tag{B.2}$$

with,

$$\begin{aligned} \alpha_2^V &:= I(V'|V) - I(V|V') \\ \alpha_3^{X \to Z} &:= \Delta I(X|Z \cup Z') \\ \alpha_5^V &:= \Delta^2 I(V) \\ \alpha_6^{X \to Z} &:= \Delta I(X'|Z \cup Z'). \end{aligned} \tag{B.3}$$

The first line is the definition of E repeated for reference. The second line decomposes the time increment of co-information into signed transfers across ordered bipartitions and a residual. The remaining lines express increments of the multiple-transfer terms through co-information increments and conditional-entropy residuals. These are identities for consistent joint laws, without a stationarity assumption; imposing that selected residuals vanish is an additional modeling restriction. Equation (15) gives the underlying recursion in the number of arguments. The residual hypotheses in Appendix C specify when this hierarchy preserves the required higher-order cancellation through time.

Appendix C. The Proof of Proposition 3

For reference, the full cycle/source balances summarized in the proposition are

$$
\begin{aligned}
&\Delta H(\Gamma_i) = T_{\Gamma_{i-1}\to\Gamma_i} - T^{\dagger}_{\Gamma_{i+1}\to\Gamma_i} \quad (i \neq p)\\
&\Delta H(\Gamma_p) = T_{\Gamma_{p-1}\to\Gamma_p} + T_{\Gamma_{Ext}\to\Gamma_p} - T^{\dagger}_{\Gamma_{p+1}\to\Gamma_p}\\
&\qquad\qquad - E_{\{\Gamma_{p-1},\Gamma_{p+1},\Gamma_{Ext}\}\to\Gamma_p}\\
&\Delta I(\Gamma_i,\Gamma_{i+1}) = T_{\Gamma_i\to\Gamma_{i+1}} - T^{\dagger}_{\Gamma_{i+1}\to\Gamma_i}\\
&\Delta I(\Gamma_p,\Gamma_{Ext}) = T_{\Gamma_{Ext}\to\Gamma_p} - T^{\dagger}_{\Gamma_p\to\Gamma_{Ext}}
\end{aligned}
$$

$$
\begin{aligned}
&\Delta T_{\Gamma_i\to\Gamma_{i+1}} = T_{\Gamma_{i-1}\to\Gamma_i} - T_{\Gamma_i\to\Gamma_{i+1}}\\
&\qquad\qquad\qquad\qquad (i \neq p-1, p)\\
&\Delta T^{\dagger}_{\Gamma_i\to\Gamma_{i+1}} = 0\\
&\Delta T_{\Gamma_i\to\Gamma_{i-1}} = 0\\
&\Delta T^{\dagger}_{\Gamma_i\to\Gamma_{i-1}} = T^{\dagger}_{\Gamma_{i-1}\to\Gamma_{i-2}} - T^{\dagger}_{\Gamma_i\to\Gamma_{i-1}}\\
&\qquad\qquad\qquad\qquad (i \neq p, p+1)
\end{aligned}
$$

$$
\begin{aligned}
&\Delta T_{\Gamma_{p-1}\to\Gamma_p} = T_{\Gamma_{p-2}\to\Gamma_{p-1}} - T_{\Gamma_{p-1}\to\Gamma_p}\\
&\qquad\qquad - \alpha_3^{\Gamma_{p-1}\to\Gamma_p}\\
&\Delta T^{\dagger}_{\Gamma_p\to\Gamma_{p-1}} = T^{\dagger}_{\Gamma_{p-1}\to\Gamma_{p-2}} - T^{\dagger}_{\Gamma_p\to\Gamma_{p-1}}\\
&\qquad\qquad - E'_{\{\Gamma_{p-1},\Gamma_{p+1},\Gamma_{Ext}\}\to\Gamma_p}\\
&\Delta T_{\Gamma_p\to\Gamma_{p+1}} = T_{\Gamma_{p-1}\to\Gamma_p} + T_{\Gamma_{Ext}\to\Gamma_p}\\
&\qquad\qquad - T_{\Gamma_p\to\Gamma_{p+1}} - E_{\{\Gamma_{p-1},\Gamma_{p+1},\Gamma_{Ext}\}\to\Gamma_p}\\
&\Delta T^{\dagger}_{\Gamma_{p+1}\to\Gamma_p} = \alpha_1^{\{\Gamma_p\}\to\Gamma_{Ext}} - \alpha_3^{\Gamma_{Ext}\to\Gamma_p}\\
&\qquad\qquad + T^{\dagger}_{\Gamma_p\to\Gamma_{p-1}} - T^{\dagger}_{\Gamma_{p+1}\to\Gamma_p}\\
&\qquad\qquad - E'_{\{\Gamma_{p-1},\Gamma_{p+1},\Gamma_{Ext}\}\to\Gamma_p} - \alpha_6^{\Gamma_p\to\Gamma_{p-1}}
\end{aligned}
$$

$$
\begin{aligned}
&\Delta T_{\Gamma_{Ext}\to\Gamma_p} = \alpha_1^{\{\Gamma_p\}\to\Gamma_{Ext}} - T_{\Gamma_{Ext}\to\Gamma_p} - \alpha_3^{\Gamma_{Ext}\to\Gamma_p}\\
&\Delta T^{\dagger}_{\Gamma_p\to\Gamma_{Ext}} = -\alpha_{3456}^{\Gamma_p\to\Gamma_{Ext}},
\end{aligned}
\tag{C.1}
$$

under the zero-control limit,

$$\begin{aligned}
&\alpha_1^{\{\Gamma_{i-1},\Gamma_{i+1}\}\to\Gamma_i} = \alpha_2^{\Gamma_i,\Gamma_{i+1}} = 0 \quad (i \neq p)\\
&\alpha'^{\{\Gamma_{i-1},\Gamma_{i+1}\}\to\Gamma_i}_1 = \alpha'^{\Gamma_i,\Gamma_{i+1}}_2 = 0 \quad (i \neq p)\\
&\alpha_1^{\{\Gamma_{p-1},\Gamma_{p+1},\Gamma_{Ext}\}\to\Gamma_p} = \alpha'^{\{\Gamma_{p-1},\Gamma_{p+1},\Gamma_{Ext}\}\to\Gamma_p}_1\\
&\qquad = \alpha_2^{\Gamma_i,\Gamma_{i+1}} = \alpha'^{\Gamma_i,\Gamma_{i+1}}_2 = 0\\
&\alpha_3^{\Gamma_i\to\Gamma_{i+1}} = 0 \quad (i \neq p-1)\\
&\alpha_6^{\Gamma_{i-1}\to\Gamma_{i-2}} = 0 \quad (i \neq p+1)\\
&\alpha_3^{\{\Gamma_{i-1},\Gamma_{i+1}\}\to\Gamma_i} = \alpha_5^{\{\Gamma_{i-1},\Gamma_{i+1}\}}\\
&\qquad = \alpha_5^{\{\Gamma_{i-1},\Gamma_i,\Gamma_{i+1}\}} = \alpha_6^{\{\Gamma_{i-1},\Gamma_{i+1}\}\to\Gamma_i} = 0\\
&\qquad\qquad (i \neq p).
\end{aligned} \tag{C.2}$$

All zero-control conditions in Eq. (C.2) are imposed at every time step over the interval considered, and the initial conditions $E_{\{\Gamma_{i-1},\Gamma_{i+1}\}\to\Gamma_i} = 0$ for all $i \neq p$ are assumed.

We now derive this specialization from the pairwise identities.

From the intrinsic method of Ref. [30], $\alpha_{3456}^{\Gamma_j\to\Gamma_i} := -\alpha_3^{\Gamma_j\to\Gamma_i} - \alpha_4^{\Gamma_j} + \alpha_5^{\Gamma_i,\Gamma_j} + \alpha_6^{\Gamma_j\to\Gamma_i}$ can be expressed as:

$$\mathbf{x} = \mathbf{A}^{-1}\mathbf{y} - \mathbf{A}^{-1}\mathbf{b}, \tag{C.3}$$

where $\mathbf{y}, \mathbf{x}, \mathbf{b} \in \mathbb{R}^n$ and $\mathbf{A} \in \mathbb{R}^{n\times n}$ such that,

$$\begin{aligned}
(\mathbf{y})_l &= \alpha_3^{\Gamma_i\to\Gamma_{j_l}} - \alpha_6^{\Gamma_i\to\Gamma_{j_l}} - \Delta\alpha_2^{\Gamma_i,\Gamma_{j_l}}\\
(\mathbf{x})_l &= \alpha_{3456}^{\Gamma_{j_l}\to\Gamma_i}\\
(\mathbf{b})_l &= \Delta E_{K_i\to\Gamma_i} - \Delta\alpha_1^{K_i\to\Gamma_i}\\
(\mathbf{A})_{lm} &= -1 + \delta_{lm},
\end{aligned} \tag{C.4}$$

Here, the component i's neighbor set is denoted as $K_i = \{\Gamma_{j_1}, \ldots, \Gamma_{j_n}\}$ with $n > 1$. Therefore,

($i \neq p$ case)

$$\begin{aligned}
\alpha_{3456}^{\Gamma_{i-1}\to\Gamma_i} &= -\alpha_3^{\Gamma_i\to\Gamma_{i+1}} + \alpha_6^{\Gamma_i\to\Gamma_{i+1}} + \Delta E^{\{\Gamma_{i-1},\Gamma_{i+1}\}\to\Gamma_i}\\
&\quad - \Delta\alpha_1^{\Gamma_i} + \Delta\alpha_2^{\Gamma_i,\Gamma_{i+1}}\\
\alpha_{3456}^{\Gamma_{i+1}\to\Gamma_i} &= -\alpha_3^{\Gamma_i\to\Gamma_{i-1}} + \alpha_6^{\Gamma_i\to\Gamma_{i-1}} + \Delta E^{\{\Gamma_{i-1},\Gamma_{i+1}\}\to\Gamma_i}\\
&\quad - \Delta\alpha_1^{\Gamma_i} + \Delta\alpha_2^{\Gamma_i,\Gamma_{i-1}}.
\end{aligned} \tag{C.5}$$

($i = p$ case)

$$\begin{pmatrix} \alpha_{3456}^{\Gamma_{p-1}\to\Gamma_p} \\ \alpha_{3456}^{\Gamma_{p+1}\to\Gamma_p} \\ \alpha_{3456}^{\Gamma_{Ext}\to\Gamma_p} \end{pmatrix} = \frac{1}{2}\begin{pmatrix} +1 & -1 & -1 \\ -1 & +1 & -1 \\ -1 & -1 & +1 \end{pmatrix}\mathbf{y} + \frac{1}{2}\mathbf{b}, \tag{C.6}$$

where,

$$\mathbf{y} = \begin{pmatrix} \alpha_3^{\Gamma_p \to \Gamma_{p-1}} - \alpha_6^{\Gamma_p \to \Gamma_{p-1}} - \Delta\alpha_2^{\Gamma_p, \Gamma_{p-1}} \\ \alpha_3^{\Gamma_p \to \Gamma_{p+1}} - \alpha_6^{\Gamma_p \to \Gamma_{p+1}} - \Delta\alpha_2^{\Gamma_p, \Gamma_{p+1}} \\ \alpha_3^{\Gamma_p \to \Gamma_{Ext}} - \alpha_6^{\Gamma_p \to \Gamma_{Ext}} - \Delta\alpha_2^{\Gamma_p, \Gamma_{Ext}} \end{pmatrix},$$

$$\mathbf{b} = [\Delta E_{\{\Gamma_{p-1}, \Gamma_{p+1}, \Gamma_{Ext}\} \to \Gamma_p} - \Delta\alpha_1^{\{\Gamma_{p-1}, \Gamma_{p+1}, \Gamma_{Ext}\} \to \Gamma_p}] \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}. \tag{C.7}$$

Substituting these relations into the dynamical equations (Eq. (20)) and imposing the blocking flows condition, $T_{\Gamma_i \to \Gamma_{i-1}}(t) = 0$, $T^\dagger_{\Gamma_i \to \Gamma_{i+1}}(t) = 0$, $T_{\Gamma_p \to \Gamma_{Ext}}(t) = 0$, $T^\dagger_{\Gamma_{Ext} \to \Gamma_p}(t) = 0$ for $\forall t, i$, yields,

$$\begin{aligned}
\Delta T_{\Gamma_i \to \Gamma_{i+1}} &= T_{\Gamma_{i-1} \to \Gamma_i} - T_{\Gamma_i \to \Gamma_{i+1}} \\
&\quad - E_{\{\Gamma_{i-1}, \Gamma_{i+1}\} \to \Gamma_i} + \alpha_1^{\{\Gamma_{i-1}, \Gamma_{i+1}\} \to \Gamma_i} \\
&\quad - \alpha_2^{\Gamma_i, \Gamma_{i+1}} - \alpha_3^{\Gamma_i \to \Gamma_{i+1}} \quad (i \neq p), \\
\Delta T^\dagger_{\Gamma_i \to \Gamma_{i+1}} &= 0, \\
\Delta T_{\Gamma_i \to \Gamma_{i-1}} &= 0, \\
\Delta T^\dagger_{\Gamma_i \to \Gamma_{i-1}} &= T^\dagger_{\Gamma_{i-1} \to \Gamma_{i-2}} - T^\dagger_{\Gamma_i \to \Gamma_{i-1}} \\
&\quad - E'_{\{\Gamma_{i-2}, \Gamma_i\} \to \Gamma_{i-1}} + \alpha'^{\{\Gamma_{i-2}, \Gamma_i\} \to \Gamma_{i-1}}_1 \\
&\quad - \alpha'^{\Gamma_{i-1}, \Gamma_i}_2 - \alpha_6^{\Gamma_{i-1} \to \Gamma_{i-2}} \quad (i \neq p+1), \\
\Delta T_{\Gamma_p \to \Gamma_{p+1}} &= T_{\Gamma_{p-1} \to \Gamma_p} + T_{\Gamma_{Ext} \to \Gamma_p} - T_{\Gamma_p \to \Gamma_{p+1}} \\
&\quad - E_{\{\Gamma_{p-1}, \Gamma_{p+1}, \Gamma_{Ext}\} \to \Gamma_p} \\
&\quad + \alpha_1^{\{\Gamma_{p-1}, \Gamma_{p+1}, \Gamma_{Ext}\} \to \Gamma_p} - \alpha_2^{\Gamma_p, \Gamma_{p+1}} - \alpha_3^{\Gamma_p \to \Gamma_{p+1}}, \\
\Delta T^\dagger_{\Gamma_{p+1} \to \Gamma_p} &= \alpha_1^{\{\Gamma_p\} \to \Gamma_{Ext}} - \alpha_3^{\Gamma_{Ext} \to \Gamma_p} \\
&\quad + T^\dagger_{\Gamma_p \to \Gamma_{p-1}} - T^\dagger_{\Gamma_{p+1} \to \Gamma_p} - \alpha_6^{\Gamma_p \to \Gamma_{p-1}} \\
&\quad - E'_{\{\Gamma_{p-1}, \Gamma_{p+1}, \Gamma_{Ext}\} \to \Gamma_p} \\
&\quad + \alpha'^{\{\Gamma_{p-1}, \Gamma_{p+1}, \Gamma_{Ext}\} \to \Gamma_p}_1 - \alpha'^{\Gamma_p, \Gamma_{p-1}}_2, \\
\Delta T_{\Gamma_{Ext} \to \Gamma_p} &= \alpha_1^{\{\Gamma_p\} \to \Gamma_{Ext}} - T_{\Gamma_{Ext} \to \Gamma_p} - \alpha_3^{\Gamma_{Ext} \to \Gamma_p}, \\
\Delta T^\dagger_{\Gamma_p \to \Gamma_{Ext}} &= -\alpha_{3456}^{\Gamma_p \to \Gamma_{Ext}},
\end{aligned} \tag{C.8}$$

where E' denotes E evaluated at the following time step.
As Ref. [30], it is required to analyze near the zero-control limit to examine the

effect of merging. Therefore,

$$\begin{aligned}
&\alpha_1^{\{\Gamma_{i-1},\Gamma_{i+1}\}\to\Gamma_i} = \alpha_2^{\Gamma_i,\Gamma_{i+1}} = 0 \quad (i \neq p),\\
&\alpha'^{\{\Gamma_{i-1},\Gamma_{i+1}\}\to\Gamma_i}_1 = \alpha'^{\Gamma_i,\Gamma_{i+1}}_2 = 0 \quad (i \neq p),\\
&\alpha_1^{\{\Gamma_{p-1},\Gamma_{p+1},\Gamma_{Ext}\}\to\Gamma_p} = \alpha'^{\{\Gamma_{p-1},\Gamma_{p+1},\Gamma_{Ext}\}\to\Gamma_p}_1\\
&\qquad\qquad = \alpha_2^{\Gamma_i,\Gamma_{i+1}} = \alpha'^{\Gamma_i,\Gamma_{i+1}}_2 = 0,\\
&\alpha_3^{\Gamma_i\to\Gamma_{i+1}} = 0 \quad (i \neq p-1),\\
&\alpha_6^{\Gamma_{i-1}\to\Gamma_{i-2}} = 0 \quad (i \neq p+1),\\
&\alpha_3^{\{\Gamma_{i-1},\Gamma_{i+1}\}\to\Gamma_i} = \alpha_5^{\{\Gamma_{i-1},\Gamma_{i+1}\}}\\
&\qquad = \alpha_5^{\{\Gamma_{i-1},\Gamma_i,\Gamma_{i+1}\}} = \alpha_6^{\{\Gamma_{i-1},\Gamma_{i+1}\}\to\Gamma_i} = 0\\
&\qquad\qquad (i \neq p),
\end{aligned} \tag{C.9}$$

retaining only the control variables associated with the newly opened channel between Ext and p. In this limit,

$$\begin{aligned}
\Delta E = &-\alpha_3^{\{\Gamma_{i-1},\Gamma_{i+1}\}\to\Gamma_i} - \alpha_5^{\{\Gamma_{i-1},\Gamma_{i+1}\}}\\
&+\alpha_5^{\{\Gamma_{i-1},\Gamma_i,\Gamma_{i+1}\}} + \alpha_6^{\{\Gamma_{i-1},\Gamma_{i+1}\}\to\Gamma_i} = 0.
\end{aligned} \tag{C.10}$$

as the high-order control variables also vanish. Together with the stated initial condition, this implies $E_{\{\Gamma_{i-1},\Gamma_{i+1}\}\to\Gamma_i} = E'_{\{\Gamma_{i-1},\Gamma_{i+1}\}\to\Gamma_i} = 0$ for $i \neq p$ throughout the interval considered.

This yields Eq. (C.1).

■

The equation (C.1) requires further specification. Firstly, $\alpha_{3456}^{\Gamma_p\to\Gamma_{Ext}}$ cannot be replaced by α_3, α_6 by the intrinsic method introduced in Ref. [30], since the component Ext has only one neighbor. Therefore, $\alpha_{3456}^{\Gamma_p\to\Gamma_{Ext}}$ has to be determined from its definition, $\alpha_{3456}^{\Gamma_p\to\Gamma_{Ext}} = -\alpha_3^{\Gamma_p\to\Gamma_{Ext}} - \alpha_4^{\Gamma_p} + \alpha_5^{\Gamma_p,\Gamma_{Ext}} + \alpha_6^{\Gamma_p\to\Gamma_{Ext}}$. Secondly, defining the dynamic characteristics of the component Ext can specify the terms, $\alpha_5^{\Gamma_p,\Gamma_{Ext}}$ and $\alpha_1^{\{\Gamma_p\}\to\Gamma_{Ext}}$. For instance, $\alpha_5^{\Gamma_p,\Gamma_{Ext}} = 0$ at all times and $\alpha_1^{\{\Gamma_p\}\to\Gamma_{Ext}} = 0$ at $t < t_1$ if the component Ext is regarded as a source of white noise.

On the other hand, if $E_{\{\Gamma_{p-1},\Gamma_{p+1},\Gamma_{Ext}\}\to\Gamma_p} = 0$ permanently, there is no stationary state with nonzero $T_{\Gamma_{Ext}\to\Gamma_p}, T^{\dagger}_{\Gamma_p\to\Gamma_{Ext}}$. This shows that multivariate effects are essential for the existence of a stationary state with nonzero interaction with the external source.

Such transient effects are expected to appear in explicit dynamical realizations of the merging process. The transient behaviors after time $t = t_1$ and the way to approach another stationary state were demonstrated numerically in Ref. [29].

Appendix D. The Proof of Proposition 4

By the definition of E,

$$\begin{aligned} E_{\{A',B',Q'\}\to P'} &= [T_{\{A',B'\}\to\{P'\}} - T^{\dagger}_{\{A',B'\}\to\{P'\}}] \\ &\quad + [T_{\{A',Q'\}\to\{P'\}} - T^{\dagger}_{\{A',Q'\}\to\{P'\}}] \\ &\quad + [T_{\{B',Q'\}\to\{P'\}} - T^{\dagger}_{\{B',Q'\}\to\{P'\}}] \\ &\quad - [T_{\{A',B',Q'\}\to\{P'\}} - T^{\dagger}_{\{A',B',Q'\}\to\{P'\}}]. \end{aligned} \tag{D.1}$$

From Assumption 5, one obtains,

$$\begin{aligned} T^{\dagger}_{\{A',Q'\}\to\{P'\}} &= I(\{A'',Q'',P'\}|P'') \\ &= I(A'',Q''|P'') - I(A'',Q''|P'P'') \\ &= I(A'',P'|P'') \quad \text{(Assumption 5: } Q''=P') \\ &= T^{\dagger}_{A'\to p'} = 0 \quad \text{(blocking-flows condition)}, \end{aligned} \tag{D.2}$$

and $T^{\dagger}_{\{B',Q'\}\to\{P'\}} = 0$ similarly. In addition,

$$\begin{aligned} &T^{\dagger}_{\{A',B',Q'\}\to\{P'\}} \\ &\quad = I(\{A'',B'',Q'',P'\}|P'') \\ &\quad = I(\{A'',B'',P'\}|P'') \\ &- I(\{A'',B'',P'\}|P''Q'') \\ &\quad = I(\{A'',B'',P'\}|P'') \quad \text{(Assumption 5: } Q''=P') \\ &\quad = T^{\dagger}_{\{A',B'\}\to\{P'\}}. \end{aligned} \tag{D.3}$$

The $T_{\{A',B',Q'\}\to\{P'\}}$ is evaluated using Assumption 2:

$$\begin{aligned} &T_{\{A',B',Q'\}\to\{P'\}} \\ &\quad = I(\{A',B',Q',P''\}|P') \\ &\quad = I(\{A',Q',P''\}|P') - I(\{A',Q',P''\}|P'B') \\ &\quad = T_{\{A',Q'\}\to\{P'\}} - I(Q',P''|P'B') \\ &+ I(Q',P''|P'A'B') \\ &\quad = T_{\{A',Q'\}\to\{P'\}} + [I(Q',P''|P') - I(Q',P''|P'B')] \\ &+ I(Q',P''|P'A'B') \\ &\quad (I(Q',P''|P') = T_{Q'\to P'} = 0) \\ &\quad = T_{\{A',Q'\}\to\{P'\}} + T_{\{B',Q'\}\to\{P'\}}, \\ &\quad \text{since, by Assumption 2,} \\ &\quad I(Q',P''|P'A'B') \\ &\qquad = H(P''|P'A'B') - H(P''|P'A'B'Q') = 0. \end{aligned} \tag{D.4}$$

Combining the preceding results, the Eq. (D.1) becomes,

$$E_{\{A',B',Q'\}\to P'} = T_{\{A',B'\}\to\{P'\}}. \tag{D.5}$$

■

Appendix E. The Proof of Lemma 1

Under Assumptions 2 and 5,

$$\begin{aligned}\alpha_1^{\{A,Q\}\to P} &= H(P'|PAQ) - H(P|P'A'Q') \\ &= H(P'|PA), \\ \alpha_1^{\{A',B',Q'\}\to P'} & \\ &= H(P''|P'A'B'Q') \\ &- H(P'|P''A''B''Q'') = H(P''|P'A'B').\end{aligned} \tag{E.1}$$

Since,

$$\begin{aligned}H(P'|P) &= T_{A\to P} + H(P'|PA) \\ &= T_{A\to P} + \alpha_1^{\{A,Q\}\to P},\end{aligned} \tag{E.2}$$

$$\begin{aligned}H(P''|P') &= T_{B'\to P'} + T_{A'\to P'} - T_{\{A',B'\}\to\{P'\}} \\ &\quad + H(P''|P'A'B') \\ &= T_{B'\to P'} + T_{A\to P} + H(P''|P'A'B') \\ &\qquad\qquad \text{(Assumption 1)} \\ &= T_{B'\to P'} + T_{A\to P} + H(P'|PA) \\ &\qquad\qquad \text{(Assumption 3)} \\ &= T_{B'\to P'} + H(P'|P).\end{aligned} \tag{E.3}$$

■

Appendix F. The Proof of Proposition 5

(a)

From Eq. (E.2),

$$\begin{aligned}H(P'|P) &= T_{A\to P} + \alpha_1^{\{A,Q\}\to P} \\ &= C_P - \epsilon,\end{aligned} \tag{F.1}$$

one obtains,

$$H(P') \geqslant H(P'|P) = C_P - \epsilon. \tag{F.2}$$

(b)

From Assumption 5,

$$\begin{aligned}T^{\dagger}_{Q'\to P'} &= H(Q''|P'') - H(Q''|P'P'') \\ &= H(P'|P'') = -\Delta H(P') + H(P''|P').\end{aligned} \tag{F.3}$$

As a result,

$$H(P''|P') = C_P - \delta. \tag{F.4}$$

Since $H(P''|P') \leqslant H(P'') \leqslant C_P$, one obtains $0 \leqslant \delta$.
Here, $\delta \leqslant \epsilon$ is derived from the non-negativity of $T_{B'\to P'}$.

(c)

From Assumption 6 and Lemma 1,

$$\begin{aligned}\Delta I(P', P) &= \Delta H(P') - \Delta H(P'|P) \\ &= \Delta H(P') - T_{B'\to P'} \geqslant -T_{B'\to P'}.\end{aligned} \tag{F.5}$$

Therefore,

$$\Delta H(P') \geqslant 0. \tag{F.6}$$

Then,

$$C_P \geqslant H(P'') \geqslant H(P') \geqslant C_P - \epsilon. \tag{F.7}$$

Hence,

$$\Delta H(P') \leqslant C_P - (C_P - \epsilon) = \epsilon. \tag{F.8}$$

■

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